# Supporting Information for BeeQ: Evaluating Quantum Fidelity Kernels as a New Approach Methodology for Molecular Toxicity Prediction

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This supplement provides the full evaluation specification, extended results, molecular-support diagnostics, and reproducibility records accompanying the main paper. The primary numerical source is the frozen evaluation campaign `BEEQ_FINAL_NESTED_STRUCT_IMPL_20260827T144143Z`. All predictive tables are read directly from its retained CSVs and checked against its molecule-level predictions. Supplementary analyses of descriptors, projections, and Gram geometry are regenerated from the four hash-matched frozen input CSVs; they are explicitly post hoc and do not change any trained model or threshold. Exploratory history is reported separately.

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## S1. Dataset, Endpoint, and Evidence Provenance

### A. Curated molecular domain

ApisTox provides 1,035 standardized compounds with acute honey-bee toxicity labels [1]. The BeeQ working domain comprises 893 curated organic molecules: 622 non-toxic and 271 toxic. The curation record documents a metal-free organic domain, structure-aware partitions, unique exported IDs, and finite X10 coordinates. The modeling code starts from these curated tables rather than re-running the complete upstream curation procedure. Consequently, 142 source compounds are excluded in total, but a sequential per-operation removal ledger cannot be reconstructed from the retained artifacts. Separate counts for salt handling, metal exclusions, failed structures, or duplicate removal are not inferred. This distinction follows the need to audit chemical curation [2], [3].

TABLE S1
Dataset composition. Cluster counts use the frozen `BUTINA_CLUSTER_ID` assignments. CR8 does not receive those cluster IDs.

| Partition | Molecules | Non-toxic | Toxic | Toxic fraction | Clusters |
|---|---|---|---|---|---|
| Development | 712 | 490 | 222 | 0.3118 | 374 |
| Historical holdout | 181 | 132 | 49 | 0.2707 | 88 |
| BeeQ893 total | 893 | 622 | 271 | 0.3035 | 462 |
| External CR8 | 8 | 6 | 2 | 0.2500 | — |

### B. Endpoint and external identities

The target is a binary class rather than a fitted continuous lethal dose:

$$y = \begin{cases} 1, & \mathrm{LD}_{50} \leq 11\,\mu\mathrm{g/bee}, \\ 0, & \mathrm{LD}_{50} > 11\,\mu\mathrm{g/bee}. \end{cases} \tag{S1}$$

The label threshold is fixed before modeling, following the ApisTox definition and bee-risk guidance [1], [4]. ApisTox aggregates the strongest available acute toxicity across exposure routes; the dataset therefore does not represent one homogeneous contact-exposure protocol. The historical holdout was inspected during earlier exploratory work. Its later post-freeze evaluation establishes separation from the current fit, while its historical status remains unchanged.

CR8 comprises eight active ingredients of Costa Rican regulatory relevance, identified in the register associated with Decree No. 43838–MAG–S–MINAE [5]. It is evaluated after the Development freeze. The frozen external CSV records identities, molecular coordinates, and labels but does not include the original quantitative $\mathrm{LD}_{50}$ measurements, exposure-route annotations, or their complete compound-by-compound source trail. Table S2 reports the available identities and labels without inventing those missing measurements. The main paper cites individual sources for the two toxic ingredients [6], [7].

TABLE S2
Identities and selected frozen descriptors of CR8. Class 1 is toxic. The input files contain labels rather than quantitative LD50 measurements.

| Compound | CAS | Class | nOP | logP | MW | TPSA |
|---|---|---|---|---|---|---|
| Isotianil | 224049-04-1 | 0 | 0 | 3.574 | 298.154 | 94.02 |
| Fluoxapiprolin | 1360819-11-9 | 0 | 0 | 5.479 | 650.076 | 152.60 |
| Ethiprole | 181587-01-9 | 1 | 0 | 3.779 | 397.209 | 90.69 |
| Cyclobutrifluram | 1460292-16-3 | 0 | 0 | 5.083 | 389.204 | 41.99 |
| Isoprothiolane | 50512-35-1 | 0 | 0 | 2.581 | 290.406 | 103.20 |
| Isocycloseram | 2061933-85-3 | 1 | 0 | 4.915 | 548.320 | 80.23 |
| Spiropidion | 1229023-00-0 | 0 | 0 | 3.709 | 422.909 | 68.31 |
| Florylpicoxamid | 1961312-55-9 | 0 | 0 | 4.176 | 512.509 | 103.82 |

### C. Source hierarchy and exploratory versions

Four final CSVs were recovered from the public BeeQ repository and verified against every SHA-256 value recorded in the frozen campaign. Their feature values are used without descriptor recalculation. The executed full-pipeline notebook contains a later run dated 2026-08-31, while the frozen campaign is dated 2026-08-27; a few quantum results differ at the fourth decimal place. Those versions are not averaged or silently interchanged. The earlier Phase 0 notebook uses the pre-correction `master.csv`; the final frozen inputs include a small number of corrected MolLogP/MolWt entries. All new EDA tables and plots below use the final frozen coordinates.

## S2. X10 Representation and Descriptor Diagnostics

### A. Ordered descriptors and preprocessing contract

Each model receives the same ordered ten-dimensional vector,

$$\mathbf{x} = (x_1, \ldots, x_{10})^{\mathsf{T}} \in \mathbb{R}^{10}. \tag{S2}$$

The descriptor order determines the corresponding quantum qubit assignment.

TABLE S3
FROZEN X10 ORDER. MW USES G/MOL; TPSA USES Å$^2$; THE COUNT DESCRIPTORS ARE DIMENSIONLESS.

| Qubit | Symbol | Stored column | Definition |
|---|---|---|---|
| 1 | logP | `MolLogP` | RDKit Wildman–Crippen octanol/water estimate [8]. |
| 2 | MW | `MolWt` | Average molecular weight. |
| 3 | TPSA | `TPSA_SP` | Topological polar surface area with sulfur/phosphorus contributions. |
| 4 | HBD | `NumHDonors` | Hydrogen-bond donor count. |
| 5 | RotB | `NumRotatableBonds` | Rotatable-bond count. |
| 6 | ArR | `NumAromaticRings` | Aromatic-ring count. |
| 7 | nHal | `nHalogen` | Number of F, Cl, Br, and I atoms. |
| 8 | nOP | `n_OP` | Count of matches to the frozen organophosphorus-like SMARTS. |
| 9 | LiPHEX | `LiPHEX_prediction` | Frozen in-house predicted $n$-hexadecane/water partition descriptor [9]. |
| 10 | fSASA | `sasa002_frac_polar_hetero_only` | Frozen fraction of solvent-accessible surface assigned to polar heteroatoms. |

The nOP SMARTS recorded in the exploratory record is `[P;X4](=[O,S])([O,S][#6])([O,S][#6])`. It counts a structural motif and does not imply that every phosphorus-containing compound is toxic. The exported count matches RDKit's number of unique substructure matches for all 893 molecules. Several distinct matches can involve the same phosphorus atom, so nOP is not simply the number of phosphorus atoms. LiPHEX supplies a partition descriptor rather than a bee-toxicity prediction. The retained records do not specify the full upstream fSASA conformer-generation, probe-radius, and surface-calculation procedure. Exact replay of the model inputs is supported by the frozen CSVs and hashes; independent regeneration of that upstream descriptor requires the original descriptor-generation code.

All features are finite and complete, so the modeling workflow performs no imputation. No outlier deletion, clipping, winsorization, synthetic sampling, or global feature selection is applied. Scale-sensitive families use `StandardScaler`; RF receives raw descriptors. For each training partition $\mathcal{T}$, the transformation is

$$z_j = \frac{x_j - \mu_{j,\mathcal{T}}}{\sigma_{j,\mathcal{T}}}, \quad \mu_{j,\mathcal{T}} = \frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}} x_{ij}, \quad \sigma^2_{j,\mathcal{T}} = \frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}} (x_{ij} - \mu_{j,\mathcal{T}})^2. \tag{S3}$$

Training-side means and population standard deviations are reused on the corresponding validation partition. Metadata such as pesticide use, source, year, CID, CAS, and name do not enter X10. PCA and UMAP are visualization tools only. The Supplementary post hoc scaler is fitted on the complete Development set, as appropriate for descriptive analyses rather than validation estimates.

### B. Distribution, univariate association, and redundancy

TABLE S4
X10 SUMMARY STATISTICS ON THE FROZEN DEVELOPMENT SET ($n = 712$); SAMPLE SD. ALL FEATURES ARE COMPLETE AND FINITE.

| Descriptor | Mean | SD | Min | Q1 | Median | Q3 | Max | Zeros |
|---|---|---|---|---|---|---|---|---|
| logP | 2.963 | 1.875 | -8.161 | 1.845 | 2.945 | 4.146 | 9.998 | 0 |
| MW | 311.076 | 111.600 | 42.041 | 229.715 | 303.581 | 376.447 | 886.133 | 0 |
| TPSA | 73.775 | 46.150 | 0.000 | 41.570 | 65.800 | 96.765 | 336.430 | 24 |
| HBD | 0.843 | 1.281 | 0.000 | 0.000 | 1.000 | 1.000 | 12.000 | 350 |
| RotB | 4.539 | 2.967 | 0.000 | 2.000 | 4.000 | 6.000 | 22.000 | 48 |
| ArR | 1.277 | 0.930 | 0.000 | 1.000 | 1.000 | 2.000 | 5.000 | 163 |
| nHal | 1.246 | 1.933 | 0.000 | 0.000 | 0.000 | 2.000 | 17.000 | 388 |
| nOP | 0.351 | 0.975 | 0.000 | 0.000 | 0.000 | 0.000 | 6.000 | 624 |
| LiPHEX | 0.303 | 1.368 | -4.225 | -0.475 | 0.202 | 1.024 | 6.554 | 0 |
| fSASA | 0.166 | 0.089 | 0.000 | 0.101 | 0.154 | 0.223 | 0.443 | 23 |

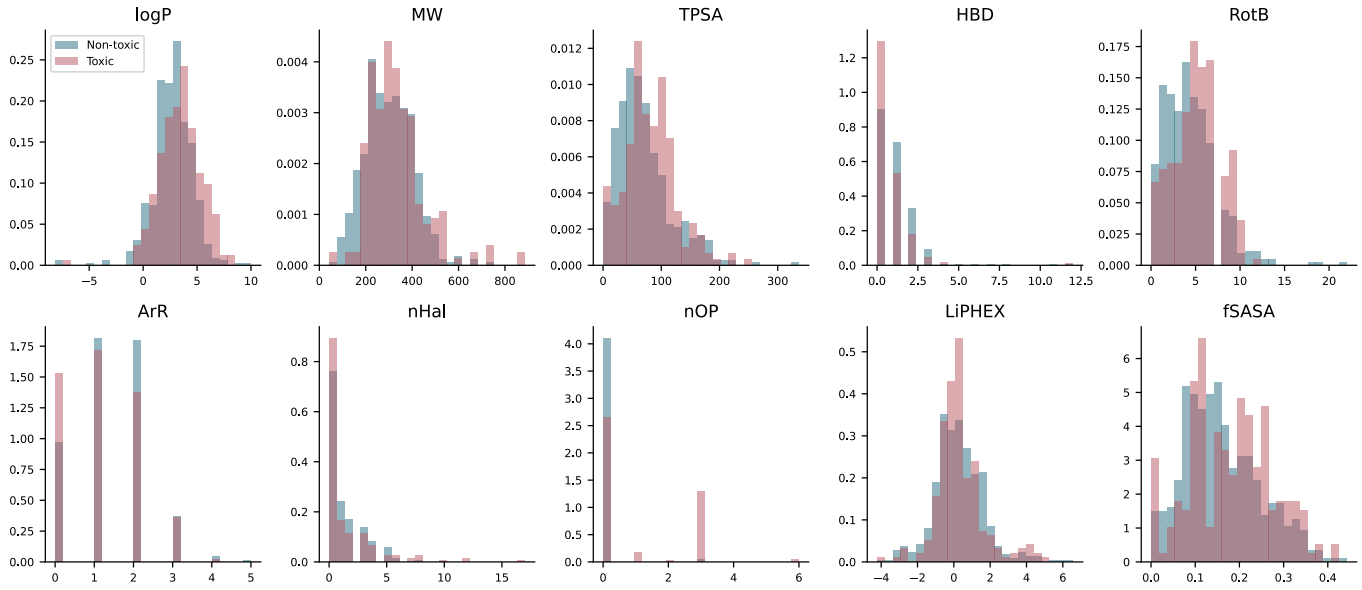



Fig. S1. Class-conditional distributions of all ten frozen Development descriptors. Histogram densities are normalized within class. Long tails and sparse count variables remain in the input representation.

TABLE S5
DEVELOPMENT-ONLY DESCRIPTIVE UNIVARIATE ASSOCIATIONS. RANK-BISERIAL SIGN IS NON-TOXIC MINUS TOXIC. BENJAMINI–HOCHBERG ADJUSTMENT COVERS THE TEN TESTS; NO DESCRIPTOR WAS SELECTED FROM THIS TABLE.

| Desc. | Med. 0 | Med. 1 | AUROC | Orient.-free | Rank biserial | $p$ | BH $q$ |
|---|---|---|---|---|---|---|---|
| logP | 2.825 | 3.507 | 0.6067 | 0.6067 | -0.2134 | 5.00e-06 | 1.67e-05 |
| MW | 298.921 | 313.794 | 0.5608 | 0.5608 | -0.1216 | 9.29e-03 | 1.16e-02 |
| TPSA | 61.405 | 75.990 | 0.5887 | 0.5887 | -0.1774 | 1.48e-04 | 2.96e-04 |
| HBD | 1.000 | 0.000 | 0.3970 | 0.6030 | 0.2060 | 1.70e-06 | 8.49e-06 |
| RotB | 4.000 | 5.000 | 0.6024 | 0.6024 | -0.2048 | 1.06e-05 | 2.64e-05 |
| ArR | 1.000 | 1.000 | 0.4338 | 0.5662 | 0.1323 | 2.93e-03 | 4.19e-03 |
| nHal | 0.000 | 0.000 | 0.4664 | 0.5336 | 0.0672 | 1.15e-01 | 1.28e-01 |
| nOP | 0.000 | 0.000 | 0.6754 | 0.6754 | -0.3507 | 1.77e-39 | 1.77e-38 |
| LiPHEX | 0.206 | 0.202 | 0.5071 | 0.5071 | -0.0142 | 7.62e-01 | 7.62e-01 |
| fSASA | 0.148 | 0.182 | 0.5731 | 0.5731 | -0.1462 | 1.77e-03 | 2.94e-03 |

TABLE S6
ORGANOPHOSPHORUS-LIKE MOTIF COUNT AND CLASS COMPOSITION. INTERVALS ARE TWO-SIDED CLOPPER–PEARSON BINOMIAL INTERVALS AND IGNORE CLUSTER DEPENDENCE; THEY ARE DESCRIPTIVE.

| Partition | nOP | $n$ | Toxic | Toxic fraction | Exact 95% low | High |
|---|---|---|---|---|---|---|
| Development | 0 | 624 | 141 | 0.2260 | 0.1937 | 0.2608 |
| Development | 1 | 9 | 9 | 1.0000 | 0.6637 | 1 |
| Development | 2 | 2 | 1 | 0.5000 | 0.0126 | 0.9874 |
| Development | 3 | 75 | 69 | 0.9200 | 0.8340 | 0.9701 |
| Development | 6 | 2 | 2 | 1.0000 | 0.1581 | 1 |
| Holdout | 0 | 169 | 39 | 0.2308 | 0.1695 | 0.3017 |
| Holdout | 1 | 6 | 6 | 1.0000 | 0.5407 | 1 |
| Holdout | 3 | 5 | 4 | 0.8000 | 0.2836 | 0.9949 |
| Holdout | 6 | 1 | 0 | 0.0000 | 0 | 0.9750 |

Single-descriptor AUROC is calculated in the increasing-feature direction; orientation-free AUROC uses $\max(A, 1 - A)$. These summaries inspect associations within Development and do not constitute cross-validated predictive results. Distributional tests use two-sided Mann–Whitney statistics with Benjamini–Hochberg adjustment over ten tests. Dependence between chemically related molecules limits the interpretation of conventional molecule-wise $p$-values. In particular, the rare nOP-positive groups require attention to their counts rather than their percentages alone.

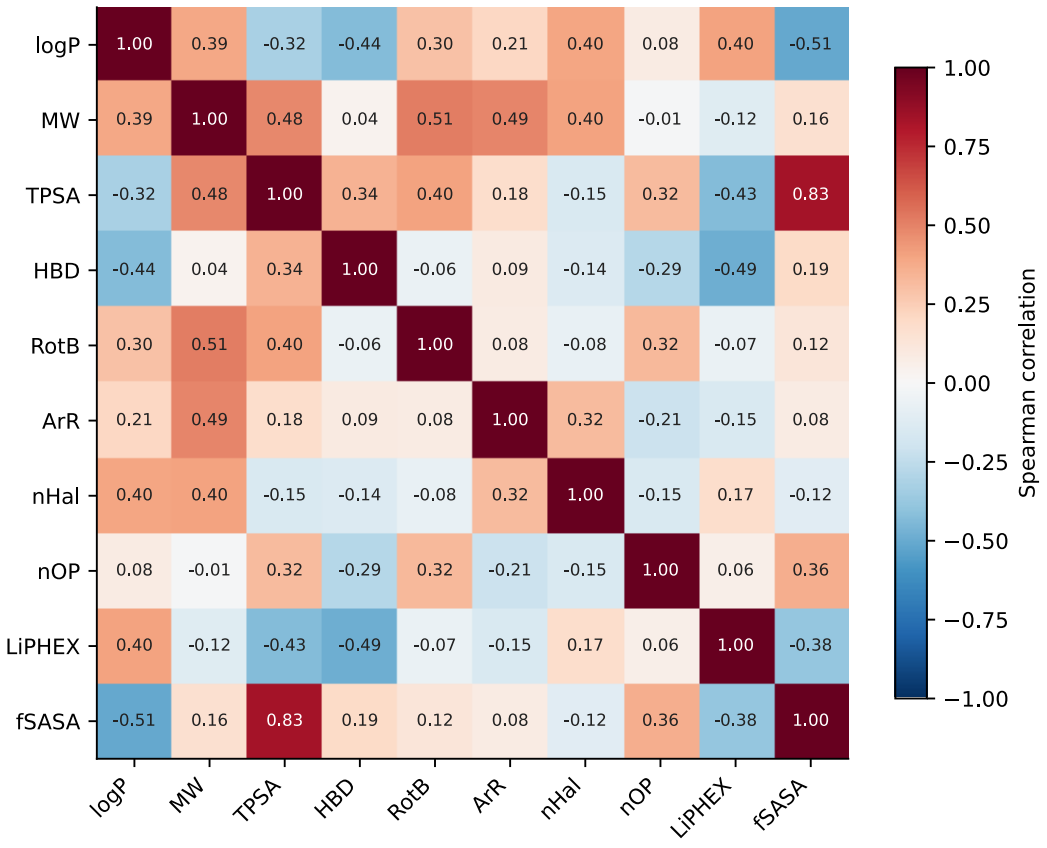



Fig. S2. Spearman correlation on the frozen Development coordinates. TPSA and polar surface fraction are strongly correlated; correlation is reported as a representation diagnostic and does not trigger feature deletion.

### *C. Linear and nonlinear projections*

TABLE S7
PCA ON STANDARDIZED FROZEN DEVELOPMENT DESCRIPTORS. COMPONENT SIGNS ARE ARBITRARY; LABEL ASSOCIATIONS ARE POST HOC. PCA IS NEVER USED AS A MODEL INPUT.

| PC | Variance | Cumulative | Orientation-free AUROC |
|---|---|---|---|
| 1 | 0.3196 | 0.3196 | 0.5481 |
| 2 | 0.2146 | 0.5341 | 0.5880 |
| 3 | 0.1564 | 0.6906 | 0.6566 |
| 4 | 0.0974 | 0.7879 | 0.5728 |
| 5 | 0.0820 | 0.8699 | 0.5532 |
| 6 | 0.0484 | 0.9183 | 0.6188 |
| 7 | 0.0378 | 0.9561 | 0.5893 |
| 8 | 0.0270 | 0.9831 | 0.6016 |
| 9 | 0.0138 | 0.9969 | 0.5108 |
| 10 | 0.0031 | 1.0000 | 0.5141 |

TABLE S8
FIRST THREE PCA LOADING COEFFICIENTS AND MUTUAL-INFORMATION ESTIMATES (SEED 42; MIXED DISCRETE/CONTINUOUS FEATURE MASK). MI IS ESTIMATOR-DEPENDENT.

| Descriptor | PC1 | PC2 | PC3 | MI |
|---|---|---|---|---|
| logP | -0.3997 | 0.3953 | 0.1183 | 0.0008 |
| MW | 0.1373 | 0.6027 | -0.1178 | 0.0262 |
| TPSA | 0.5059 | 0.2179 | 0.0881 | 0.0985 |
| HBD | 0.3871 | -0.0442 | -0.3196 | 0.0213 |
| RotB | 0.0947 | 0.3950 | 0.3822 | 0.0303 |
| ArR | 0.0394 | 0.3747 | -0.3371 | 0.0089 |
| nHal | -0.1951 | 0.3564 | -0.2653 | 0.0163 |
| nOP | 0.0843 | 0.0689 | 0.6758 | 0.1209 |
| LiPHEX | -0.4150 | 0.0452 | 0.1852 | 0.0127 |
| fSASA | 0.4333 | 0.0287 | 0.2027 | 0.0000 |

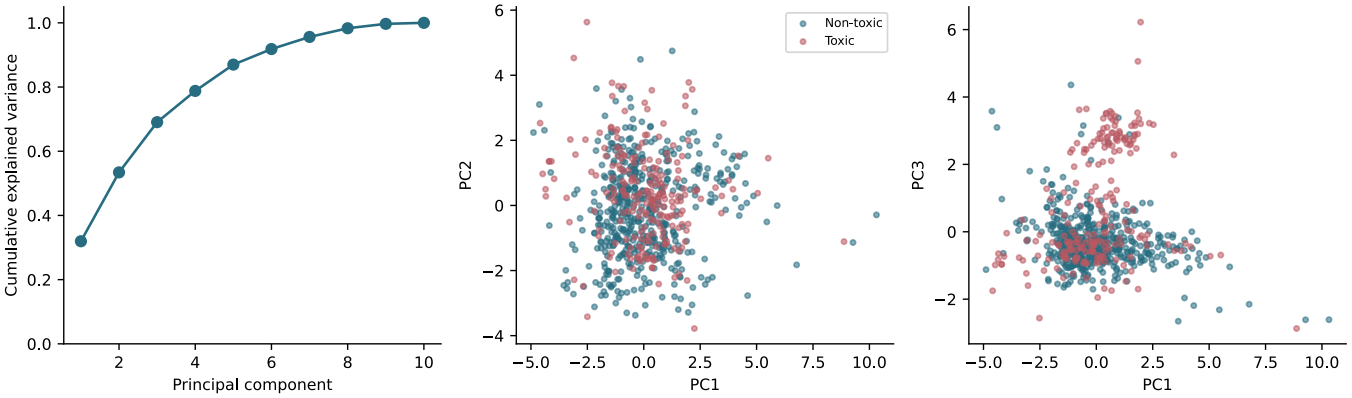



Fig. S3. PCA of standardized Development X10: cumulative variance, PC1–PC2, and PC1–PC3 colored by toxicity. Variance explained and label separation measure different properties; component signs are arbitrary.

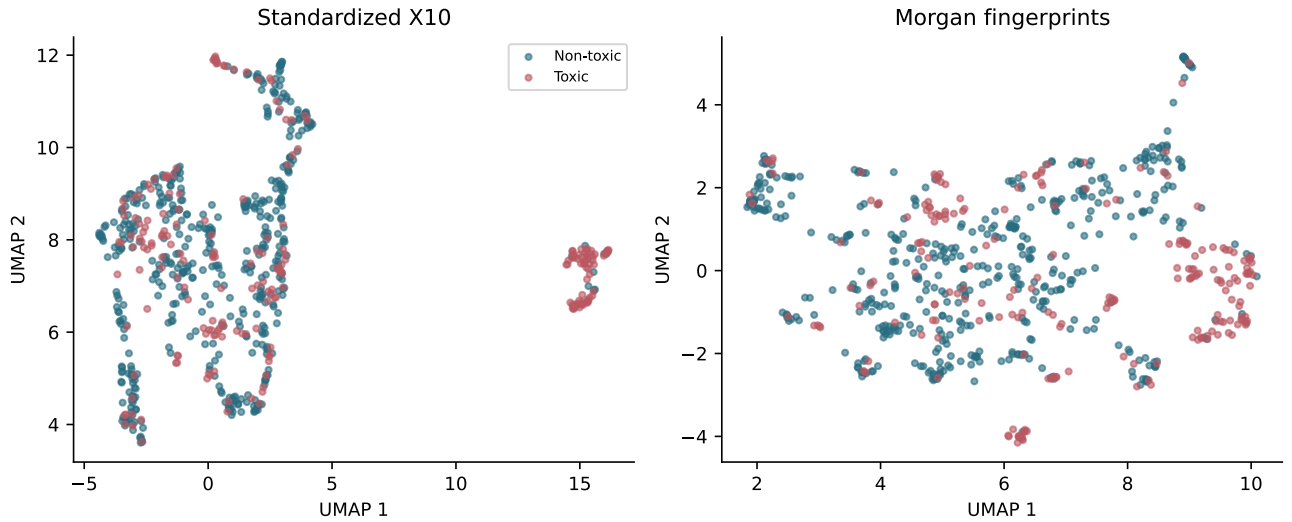



Fig. S4. Post hoc Development UMAP projections using the final frozen inputs: Euclidean distance on standardized X10 and Jaccard distance on Morgan fingerprints. Both use 15 neighbors, minimum distance 0.1, and seed 42. Embedding axes are arbitrary. Visible separation is parameter-dependent and is not evidence of predictive generalization.

The molecular comparison uses all ten original descriptors, irrespective of their low-dimensional appearance in the projections.

## S3. Structure-Aware Partitions and Nested Evaluation

### A. Fingerprints and frozen clusters

Each molecular structure is encoded with a 1,024-bit Morgan fingerprint of radius 2 with chirality enabled [10], [11]. For binary fingerprints $A$ and $B$,

$$T(A,B) = \frac{|A \cap B|}{|A \cup B|}, \qquad d(A,B) = 1 - T(A,B). \tag{S4}$$

Butina clustering uses distance cutoff 0.60 [12]. This is a neighbor-to-cluster-center threshold, equivalent to $T \geq 0.40$ in that assignment rule. It does not enforce a universal maximum similarity between every pair of molecules in different clusters. The frozen whole dataset has 462 clusters, including 334 singletons; the largest contains 40 molecules. Complete clusters are allocated to Development or the historical holdout, with no shared IDs. These are structure-aware partitions; cluster disjointness alone is not a formal out-of-distribution guarantee.

TABLE S9
FROZEN OUTER-FOLD PARTITION AUDIT. NO CLUSTER OCCURS IN BOTH SIDES OF AN OUTER FOLD. THE RETAINED CAMPAIGN ALSO CHECKS ALL INNER SPLITS.

| Fold | Train | Valid | Valid 0 | Valid 1 | Train clusters | Valid clusters | Overlap |
|---|---|---|---|---|---|---|---|
| 1 | 575 | 137 | 97 | 40 | 299 | 75 | 0 |
| 2 | 607 | 105 | 68 | 37 | 304 | 70 | 0 |
| 3 | 565 | 147 | 110 | 37 | 295 | 79 | 0 |
| 4 | 531 | 181 | 122 | 59 | 300 | 74 | 0 |
| 5 | 570 | 142 | 93 | 49 | 298 | 76 | 0 |

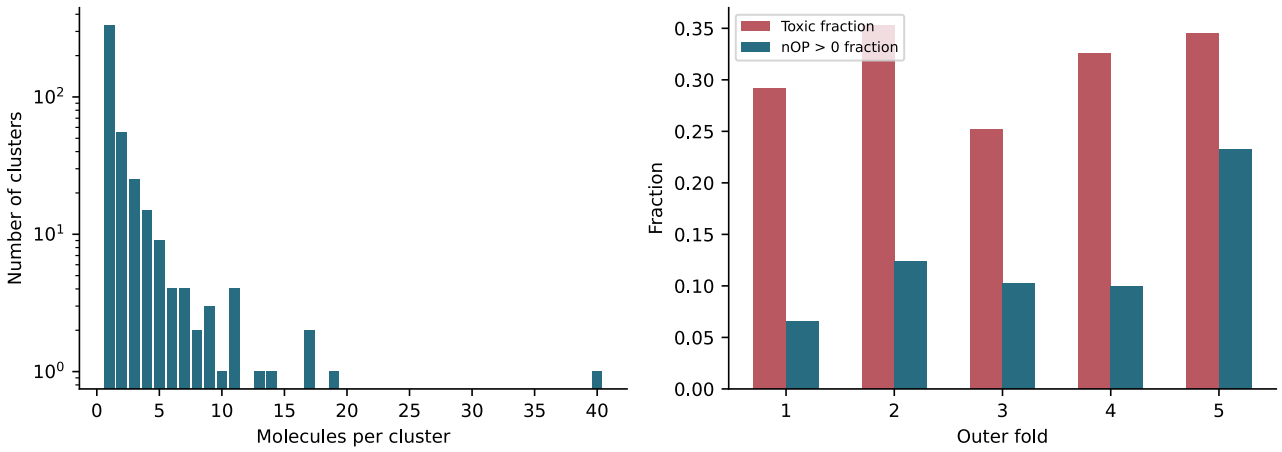



Fig. S5. Cluster-size frequencies across BeeQ893 and outer-fold class/motif composition in Development. The cluster-frequency axis is logarithmic to expose the strong singleton contribution.

TABLE S10
POST HOC CLASS PURITY WEIGHTED BY MOLECULES IN NON-SINGLETON CLUSTERS. LABELS ARE GLOBALLY PERMUTED 2,000 TIMES WITHIN EACH LISTED SUBSET (SEED 42); PLUS-ONE TAIL FRACTION. THIS IS A STRUCTURAL DIAGNOSTIC, NOT MODEL VALIDATION.

| Subset | $n$ | Non-singleton groups | Purity | Null mean | Null P95 | $p_{+1}$ |
|---|---|---|---|---|---|---|
| All Development | 712 | 102 | 0.8977 | 0.7373 | 0.7614 | 0.0005 |
| nOP = 0 | 624 | 86 | 0.8940 | 0.7968 | 0.8207 | 0.0005 |

### B. Training, selection, and threshold flow

Five stored `STRICT_CV_FOLD` assignments define the outer validation partitions. Within each outer-training partition, four shuffled deterministic `StratifiedGroupKFold` splits use `BUTINA_CLUSTER_ID`, with seed 20260824+ outer-fold index. Inner-fold mean AUROC selects hyperparameters. Preprocessing is refitted inside each permitted training split. The stored split audit records zero train/validation cluster intersections, supporting the nested design [13], [14].

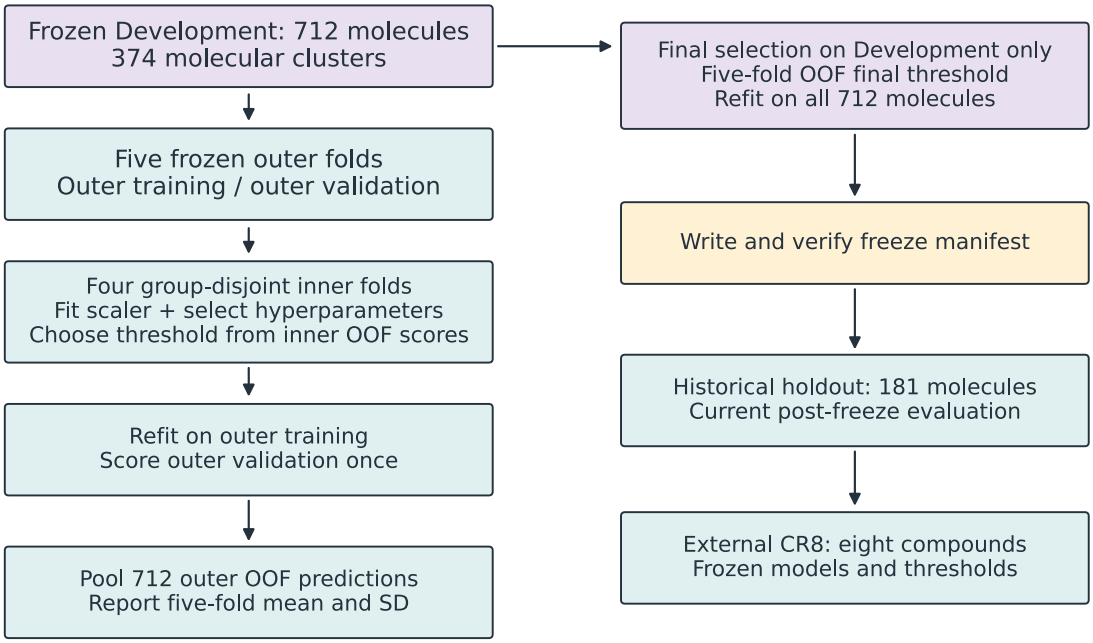



Fig. S6. Training-side selection and downstream evaluation. The outer-validation scores estimate the full selection procedure. Final Development-only selection produces a separate refitted model and frozen threshold for each family. Historical-holdout status reflects earlier inspection.

Let $f_i$ be the inner-OOF score and $t$ a candidate threshold. Positive predictions use a greater-than-or-equal comparison:

$$\widehat{y}_i(t) = \mathbb{1}\{f_i \geq t\}, \qquad t^* = \arg\min_{t \in \mathcal{C}} \left(-\mathrm{MCC}(t), -\mathrm{BA}(t), |t|, t\right) \tag{S5}$$

with lexicographic ordering. Candidate set $\mathcal{C}$ contains every unique OOF score and the next floating-point number above the largest score. Thus all-positive and all-negative possibilities are included. LR, RF, and MLP provide class-1 probabilities; the three kernel SVCs provide decision scores. Neither their threshold values nor their margin magnitudes are directly comparable between families.

After final hyperparameter selection on Development alone, five-fold Development OOF scores determine the final threshold, followed by a single refit on all 712 Development molecules. The manifest is written before the current campaign opens the holdout and CR8. This current execution order does not erase prior historical-holdout inspection.

## S4. Classical Models and Quantum Kernel Mathematics

### A. *Fixed settings and classical reference*

LR uses `liblinear`, maximum 5,000 iterations, and balanced class weights. RF uses 300 trees, square-root feature subsampling, balanced subsample weights, and no scaling. MLP uses ReLU, Adam, native early stopping, and maximum 2,000 iterations; it has no explicit class weighting in the recorded implementation. The MLP's native stopping split is an internal random validation split rather than a group-aware stopping split; the outer and inner evaluation partitions remain cluster-disjoint.

The common kernel classifier is a balanced, precomputed-kernel SVC. For $\widetilde{y}_i = 2y_i - 1$ its dual formulation is [15]

$$\begin{aligned} \max_{\boldsymbol{\alpha}} \quad & \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j \widetilde{y}_i \widetilde{y}_j K(\mathbf{z}_i, \mathbf{z}_j), \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C w_i, \qquad \sum_i \alpha_i \widetilde{y}_i = 0, \qquad w_i = \frac{n}{2 n_{y_i}}. \end{aligned} \tag{S6}$$

Its decision function is

$$f(\mathbf{z}) = \sum_{i \in \mathrm{SV}} \alpha_i \widetilde{y}_i K(\mathbf{z}_i, \mathbf{z}) + b. \tag{S7}$$

BeeQ applies the separately estimated threshold of Eq. (S5) to this score. The classical reference is

$$K_{\mathrm{RBF}}(\mathbf{z}, \mathbf{z}') = \exp\left(-\gamma \|\mathbf{z} - \mathbf{z}'\|_2^2\right). \tag{S8}$$

"Matched RBF" means matched descriptors, fold-local scaling, SVC formulation, and partitions; its selected bandwidth is allowed to differ from the quantum encoding scale.

### B. States, amplitudes, and fidelity

The ten-qubit computational basis has dimension $2^{10} = 1,024$. A normalized state associates complex amplitudes with bit strings $b$:

$$|\phi(\mathbf{z})\rangle = \sum_{b\in\{0,1\}^{10}} a_b(\mathbf{z})|b\rangle, \qquad \sum_b |a_b(\mathbf{z})|^2 = 1. \tag{S9}$$

Born probabilities are $|a_b|^2$; the present simulations use the amplitudes directly without drawing measurement shots. Fidelity defines the molecular similarity [16], [17]:

$$K_Q(\mathbf{z},\mathbf{z}') = |\langle\phi(\mathbf{z})|\phi(\mathbf{z}')\rangle|^2 = \mathrm{Tr}[\rho(\mathbf{z})\rho(\mathbf{z}')], \qquad \rho(\mathbf{z}) = |\phi(\mathbf{z})\rangle\langle\phi(\mathbf{z})|. \tag{S10}$$

The trace expression is an inner product of density operators and explains the positive-semidefinite Gram matrix in exact arithmetic. It also implies $0 \le K_Q \le 1$ and $K_Q(\mathbf{z},\mathbf{z}) = 1$. High-dimensional feature states alone establish neither a new molecular geometry nor a predictive benefit [18], [19].

### C. Product-state map and exact classical closed form

With searched encoding scale $s$, the Product map is

$$|\phi_P(\mathbf{z})\rangle = \bigotimes_{k=1}^{10} R_Y(sz_k)|0\rangle, \quad R_Y(\theta) = \begin{pmatrix}\cos(\theta/2) & -\sin(\theta/2)\\ \sin(\theta/2) & \cos(\theta/2)\end{pmatrix}. \tag{S11}$$

There is no entangling operation in the recorded Product map. Its fidelity factorizes into a directly computable expression:

$$K_P(\mathbf{z},\mathbf{z}') = \prod_{k=1}^{10} \cos^2\left(\frac{s(z_k - z'_k)}{2}\right). \tag{S12}$$

The logarithmic expansion exposes the local Gaussian connection:

$$\log K_P = -\frac{s^2}{4}\sum_k (z_k - z'_k)^2 - \frac{s^4}{96}\sum_k (z_k - z'_k)^4 + O\left(s^6 \sum_k |z_k - z'_k|^6\right). \tag{S13}$$

For small encoded coordinate differences this yields

$$K_P \approx \exp\left[-\gamma_{\mathrm{eff}}\|\mathbf{z} - \mathbf{z}'\|_2^2\right], \qquad \gamma_{\mathrm{eff}} = s^2/4. \tag{S14}$$

At final $s = 0.25$, $\gamma_{\mathrm{eff}} = 0.015625$. The approximation is local: the cosine factors are periodic and their zeros invalidate a uniform global Gaussian equivalence. Product fidelity can be evaluated classically in $O(10)$ work per pair without a statevector.

### D. IQP-ZZ map and gate convention

The canonical NumPy IQP map prepares a uniform superposition and applies one diagonal phase layer. For $\sigma_k(b) = 1 - 2b_k$, its state is

$$|\phi_I(\mathbf{z})\rangle = 2^{-5} \sum_{b\in\{0,1\}^{10}} e^{i\theta_b(\mathbf{z})}|b\rangle, \quad \theta_b(\mathbf{z}) = s\sum_{k=1}^{10} z_k\sigma_k(b) + \lambda s^2 \sum_{k=1}^{9} z_k z_{k+1}\sigma_k(b)\sigma_{k+1}(b). \tag{S15}$$

The interaction strength is fixed at $\lambda = 1$, the interaction graph is a nearest-neighbor chain, and the repetition count is one. In conventional rotation notation the same unitary is

$$U_I(\mathbf{z}) = \prod_{k=1}^{9} R_{ZZ,k,k+1}(-2\lambda s^2 z_k z_{k+1}) \prod_{k=1}^{10} R_{Z,k}(-2sz_k)\, H^{\otimes 10}, \tag{S16}$$

where $R_Z(\vartheta) = e^{-i\vartheta Z/2}$ and $R_{ZZ}(\vartheta) = e^{-i\vartheta Z\otimes Z/2}$. The minus signs and factor two follow the implemented $+i\theta$ phase convention. There is no final Hadamard layer. Appending the same data-independent unitary to both feature states would preserve their fidelity.

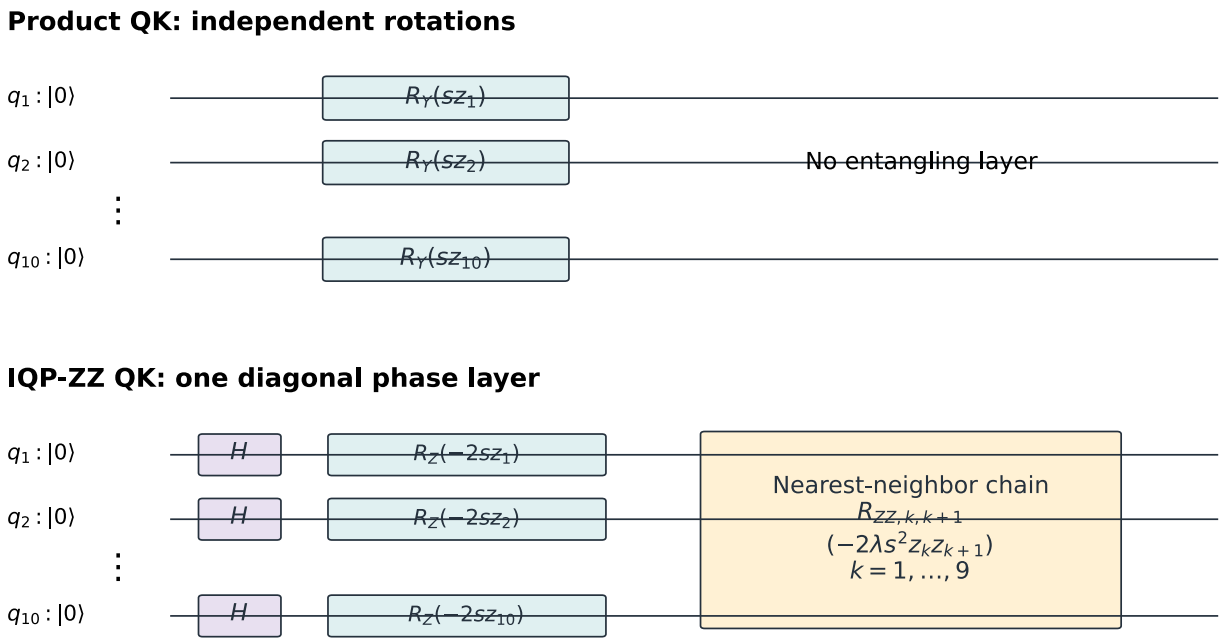



Fig. S7. Canonical feature maps. Product uses ten independent $R_Y$ rotations. IQP-ZZ uses initial Hadamards, $R_Z$ rotations, and a chain of data-dependent $R_{ZZ}$ phases with the sign convention of Eq. (S16). Ellipses denote the intervening qubits and chain couplings.

The exact IQP fidelity can also be written as a finite sum:

$$K_I(\mathbf{z}, \mathbf{z}') = \left| 2^{-10} \sum_b e^{i[\theta_b(\mathbf{z}') - \theta_b(\mathbf{z})]} \right|^2 . \tag{S17}$$

When $\lambda = 0$, this reduces to $\prod_k \cos^2[s(z_k - z'_k)]$, the Product kernel at encoding scale $2s$. Thus the selected IQP $s = 0.125$ and selected Product $s = 0.25$ have the same single-coordinate local bandwidth before IQP interactions are introduced. This provides a useful analytical context for their observed high alignment.

### *E. Simulation scope and implementation checks*

Both maps use exact noiseless NumPy statevectors, 1,024 amplitudes per molecule, without shots, a noise model, or quantum-device execution. Qiskit appears in the recorded environment but does not execute the reported campaign. For $n$ molecules, storing states requires $O(n2^{10})$ complex values and forming all pairwise overlaps requires $O(n^2 2^{10})$ arithmetic in the generic statevector route. The Product closed form reduces this cost. These results establish neither a hardware benchmark nor quantum advantage.

TABLE S11
ALL 262 RETAINED QUANTUM GRAM-MATRIX QC RECORDS, INCLUDING SEARCHED, SELECTED, AND FINALIZATION STAGES. TOLERANCES: SYMMETRY/DIAGONAL $10^{-10}$ AND MINIMUM EIGENVALUE $-10^{-8}$.

| Kernel | Matrices | Max symmetry error | Max diagonal error | Min eigenvalue | Passed |
|---|---|---|---|---|---|
| Product QK | 131 | 0.000000000000 | 3.109e-15 | -7.868e-16 | 131 |
| IQP-ZZ QK | 131 | 0.000000000000 | 8.882e-16 | -3.612e-15 | 131 |

The original campaign records that an independent Qiskit/PennyLane reference implementation was unavailable. The supplementary build additionally checks Product overlaps against Eq. (S12) and IQP amplitudes against an explicit sequence of diagonal phase multiplications on seeded synthetic coordinates. The maximum absolute errors are $1.6653 \times 10^{-16}$ for Product overlaps and $1.5206 \times 10^{-16}$ for IQP amplitudes. These checks validate the formulas and coordinate ordering without claiming a newly performed hardware experiment.

## S5. Hyperparameter Searches and Frozen Selections

TABLE S12
COMPLETE INNER SEARCH GRIDS. KERNEL FAMILIES SHARE 16 CANDIDATES; CLASSICAL SEARCH SIZES DIFFER.

| Model | Search space | Candidates |
|---|---|---|
| RF | max_depth=[None, 6], min_samples_leaf=[1, 3] | 4 |
| LR | C=[0.01, 0.1, 1.0, 10.0, 100.0] | 5 |
| Product QK | C=[0.1, 1.0, 10.0, 100.0], feature_scale=[0.125, 0.25, 0.5, 1.0] | 16 |
| IQP-ZZ QK | C=[0.1, 1.0, 10.0, 100.0], feature_scale=[0.125, 0.25, 0.5, 1.0], interaction_strength=[1.0] | 16 |
| RBF | C=[0.1, 1.0, 10.0, 100.0], gamma=[0.01, 0.03, 0.1, 0.3] | 16 |
| MLP | alpha=[0.0001, 0.001, 0.01], hidden_layer_sizes=[[32], [32, 16]], learning_rate_init=[0.0003, 0.001] | 12 |

The three kernel families each have 16 candidates, although their geometric parameters differ. LR, RF, and MLP use 5, 4, and 12 candidates respectively; the six families therefore do not have identical total search budgets. Model settings and search spaces are retained in the campaign JSON files.

TABLE S13: Independently selected outer-fold configurations; inner AUROCs are selection diagnostics, not independent validation estimates.

| Model | Fold | Parameters | Inner AUROC | Inner SD | Threshold |
|---|---|---|---|---|---|
| RF | 1 | max_depth=None, min_samples_leaf=3 | 0.7899 | 0.0424 | 0.5319 |
| RF | 2 | max_depth=None, min_samples_leaf=3 | 0.7349 | 0.0928 | 0.5602 |
| RF | 3 | max_depth=None, min_samples_leaf=1 | 0.7254 | 0.1093 | 0.6200 |
| RF | 4 | max_depth=None, min_samples_leaf=3 | 0.7491 | 0.0823 | 0.6464 |
| RF | 5 | max_depth=None, min_samples_leaf=1 | 0.7644 | 0.0518 | 0.5367 |
| LR | 1 | C=1.0 | 0.7845 | 0.0440 | 0.5761 |
| LR | 2 | C=1.0 | 0.7241 | 0.0792 | 0.6977 |
| LR | 3 | C=100.0 | 0.7264 | 0.0735 | 0.6498 |
| LR | 4 | C=1.0 | 0.7295 | 0.0586 | 0.6174 |
| LR | 5 | C=1.0 | 0.7235 | 0.0524 | 0.7022 |
| Product QK | 1 | C=1.0, feature_scale=0.5 | 0.8069 | 0.0447 | 0.0290 |
| Product QK | 2 | C=10.0, feature_scale=0.125 | 0.7468 | 0.1069 | 0.1667 |
| Product QK | 3 | C=10.0, feature_scale=0.125 | 0.7310 | 0.0824 | 0.5062 |
| Product QK | 4 | C=1.0, feature_scale=0.125 | 0.7111 | 0.1079 | -0.2696 |
| Product QK | 5 | C=10.0, feature_scale=0.125 | 0.7308 | 0.0554 | 0.3100 |
| IQP-ZZ QK | 1 | C=10.0, feature_scale=0.125, interaction_strength=1.0 | 0.8068 | 0.0431 | 0.1826 |
| IQP-ZZ QK | 2 | C=1.0, feature_scale=0.125, interaction_strength=1.0 | 0.7456 | 0.1077 | 0.1770 |
| IQP-ZZ QK | 3 | C=1.0, feature_scale=0.125, interaction_strength=1.0 | 0.7104 | 0.1182 | 0.4358 |
| IQP-ZZ QK | 4 | C=0.1, feature_scale=0.5, interaction_strength=1.0 | 0.7096 | 0.1497 | 0.0654 |
| IQP-ZZ QK | 5 | C=1.0, feature_scale=0.125, interaction_strength=1.0 | 0.7270 | 0.0537 | 0.0684 |
| RBF | 1 | C=10.0, gamma=0.03 | 0.8050 | 0.0434 | -0.0476 |
| RBF | 2 | C=1.0, gamma=0.01 | 0.7448 | 0.1010 | -0.1515 |
| RBF | 3 | C=10.0, gamma=0.01 | 0.7195 | 0.0882 | 0.9301 |
| RBF | 4 | C=1.0, gamma=0.1 | 0.7114 | 0.1277 | 0.5684 |
| RBF | 5 | C=10.0, gamma=0.01 | 0.7293 | 0.0420 | 0.2401 |
| MLP | 1 | alpha=0.01, hidden_layer_sizes=[32, 16], learning_rate_init=0.001 | 0.7199 | 0.0381 | 0.5240 |
| MLP | 2 | alpha=0.01, hidden_layer_sizes=[32, 16], learning_rate_init=0.001 | 0.6856 | 0.0516 | 0.5399 |
| MLP | 3 | alpha=0.001, hidden_layer_sizes=[32, 16], learning_rate_init=0.001 | 0.6431 | 0.1186 | 0.5206 |
| MLP | 4 | alpha=0.01, hidden_layer_sizes=[32, 16], learning_rate_init=0.001 | 0.6927 | 0.0809 | 0.5277 |
| MLP | 5 | alpha=0.0001, hidden_layer_sizes=[32, 16], learning_rate_init=0.001 | 0.6752 | 0.0326 | 0.5134 |

TABLE S14
DEVELOPMENT-ONLY FINAL SELECTIONS AND THRESHOLDS. THESE CV SCORES REUSE DEVELOPMENT AND ARE NOT INDEPENDENT PERFORMANCE ESTIMATES.

| Model | Parameters | Selection mean | SD | Threshold |
|---|---|---|---|---|
| RF | max_depth=None, min_samples_leaf=3 | 0.7501 | 0.0505 | 0.6826 |
| LR | C=100.0 | 0.7328 | 0.0930 | 0.6605 |
| Product QK | C=1.0, feature_scale=0.25 | 0.7343 | 0.1158 | 0.6462 |
| IQP-ZZ QK | C=1.0, feature_scale=0.125, interaction_strength=1.0 | 0.7346 | 0.1195 | 0.3835 |
| RBF | C=1.0, gamma=0.03 | 0.7350 | 0.1112 | 0.9157 |
| MLP | alpha=0.01, hidden_layer_sizes=[32, 16], learning_rate_init=0.0003 | 0.6863 | 0.1104 | 0.5246 |

## S6. METRIC DEFINITIONS AND EXTENDED PREDICTIVE RESULTS

### A. Thresholded and ranking metrics

With TP, FN, FP, and TN referring to toxic positives and non-toxic negatives,

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP}+\text{FN}}, \qquad \text{Specificity} = \frac{\text{TN}}{\text{TN}+\text{FP}}, \qquad \text{BA} = \frac{\text{Sensitivity}+\text{Specificity}}{2}, \tag{S18}$$

$$\text{MCC} = \frac{\text{TP}\,\text{TN} - \text{FP}\,\text{FN}}{\sqrt{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FP})(\text{TN}+\text{FN})}}. \tag{S19}$$

The retained implementation uses scikit-learn's convention of MCC zero when the denominator degenerates. AUROC is the pairwise ranking probability with half credit for tied scores. Campaign "AUPRC" is computed by `average_precision_score` [20], [21]:

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k. \tag{S20}$$

This recall-weighted step summary differs from trapezoidal interpolation of a precision–recall curve. The no-skill AP reference is the positive prevalence, which differs between Development and holdout.

For outer-fold scores $M_f$, the reported dispersion is

$$\overline{M} = \frac{1}{5}\sum_{f=1}^{5} M_f, \qquad s_M = \sqrt{\frac{1}{4}\sum_{f=1}^{5}(M_f - \overline{M})^2}. \tag{S21}$$

This is sample SD over dependent training/evaluation procedures, not a confidence interval. Unweighted mean outer AUROC/AP, pooled OOF AUROC/AP, and pooled thresholded metrics answer different questions. Scores from different fitted folds may have different scales; pooled ranking estimates should be interpreted with that limitation in mind.

### B. Development, fold-level variability, and historical transfer

TABLE S15
PRIMARY NESTED RESULTS: UNWEIGHTED MEANS AND SAMPLE SD OVER FIVE OUTER FOLDS. POOLED OOF ESTIMATES ARE DISTINCT ESTIMANDS. AP DENOTES THE CAMPAIGN AUPRC COMPUTED AS AVERAGE PRECISION.

| Model | Outer AUROC ± SD | Outer AP ± SD | Pooled AUROC | Pooled AP | BA | MCC |
|---|---|---|---|---|---|---|
| RF | 0.7480 ± 0.0574 | 0.6573 ± 0.0966 | 0.7565 | 0.6749 | 0.6798 | 0.4680 |
| LR | 0.7328 ± 0.1039 | 0.6596 ± 0.1241 | 0.7252 | 0.6454 | 0.6784 | 0.4461 |
| Product QK | 0.7216 ± 0.1075 | 0.6279 ± 0.1263 | 0.7171 | 0.5978 | 0.6576 | 0.3966 |
| IQP-ZZ QK | 0.7130 ± 0.1056 | 0.6455 ± 0.1122 | 0.7140 | 0.6151 | 0.6549 | 0.4000 |
| RBF | 0.7106 ± 0.1084 | 0.6404 ± 0.1067 | 0.7154 | 0.6260 | 0.6633 | 0.4055 |
| MLP | 0.6771 ± 0.1245 | 0.5961 ± 0.1383 | 0.6677 | 0.5763 | 0.6536 | 0.4283 |

TABLE S16
DEVELOPMENT POOLED OUTER OOF DECISIONS AT FOLD-SPECIFIC INNER-OOF THRESHOLDS.

| Model | Sens. | Spec. | TP | FN | FP | TN |
|---|---|---|---|---|---|---|
| RF | 0.3964 | 0.9633 | 88 | 134 | 18 | 472 |
| LR | 0.4099 | 0.9469 | 91 | 131 | 26 | 464 |
| Product QK | 0.3784 | 0.9367 | 84 | 138 | 31 | 459 |
| IQP-ZZ QK | 0.3649 | 0.9449 | 81 | 141 | 27 | 463 |
| RBF | 0.3919 | 0.9347 | 87 | 135 | 32 | 458 |
| MLP | 0.3378 | 0.9694 | 75 | 147 | 15 | 475 |

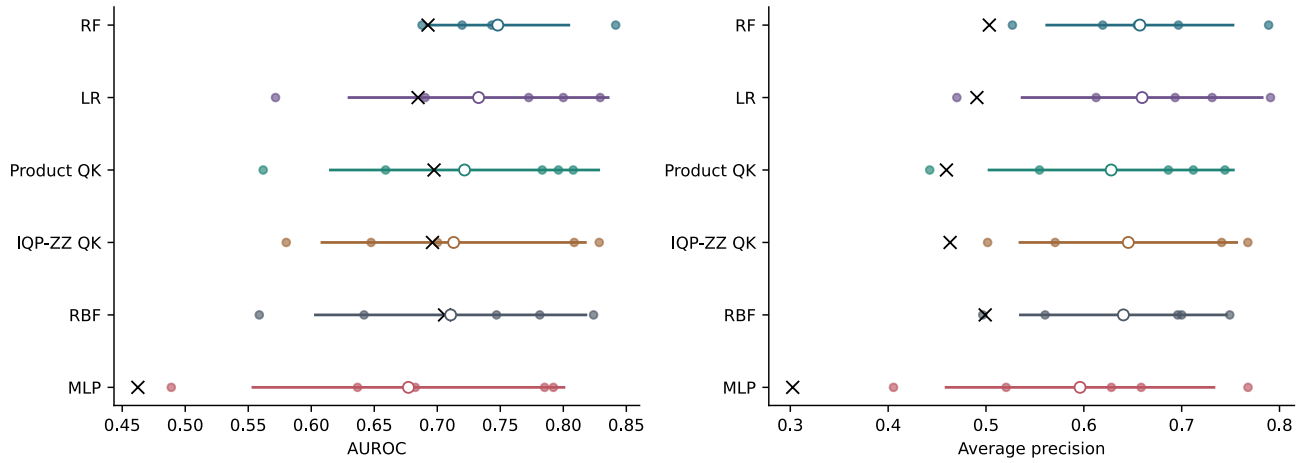



Fig. S8. Fold-level AUROC and AP, open mean markers with one sample SD, and black crosses for the historical holdout. The large fold spread is part of the structure-aware evaluation result.

TABLE S17: All 30 outer-fold results. Fold-specific thresholds and hyperparameters are listed separately.

| Model | Fold | AUROC | AP | BA | MCC | Sens. | Spec. | TP | FN | FP | TN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RF | 1 | 0.6879 | 0.5270 | 0.5963 | 0.2533 | 0.2750 | 0.9175 | 11 | 29 | 8 | 89 |
| RF | 2 | 0.7476 | 0.6546 | 0.6598 | 0.4050 | 0.3784 | 0.9412 | 14 | 23 | 4 | 64 |
| RF | 3 | 0.7197 | 0.6193 | 0.6711 | 0.5060 | 0.3514 | 0.9909 | 13 | 24 | 1 | 109 |
| RF | 4 | 0.7433 | 0.6967 | 0.6610 | 0.4925 | 0.3220 | 1.0000 | 19 | 40 | 0 | 122 |
| RF | 5 | 0.8416 | 0.7889 | 0.7894 | 0.6326 | 0.6327 | 0.9462 | 31 | 18 | 5 | 88 |
| LR | 1 | 0.5716 | 0.4702 | 0.5852 | 0.1921 | 0.3250 | 0.8454 | 13 | 27 | 15 | 82 |
| LR | 2 | 0.6904 | 0.6125 | 0.6389 | 0.3521 | 0.3514 | 0.9265 | 13 | 24 | 5 | 63 |
| LR | 3 | 0.8000 | 0.6933 | 0.6936 | 0.5255 | 0.4054 | 0.9818 | 15 | 22 | 2 | 108 |
| LR | 4 | 0.7726 | 0.7311 | 0.6654 | 0.4842 | 0.3390 | 0.9918 | 20 | 39 | 1 | 121 |
| LR | 5 | 0.8293 | 0.7908 | 0.7900 | 0.6528 | 0.6122 | 0.9677 | 30 | 19 | 3 | 90 |
| Product QK | 1 | 0.5619 | 0.4425 | 0.5425 | 0.0986 | 0.2500 | 0.8351 | 10 | 30 | 16 | 81 |
| Product QK | 2 | 0.6590 | 0.5547 | 0.6316 | 0.3265 | 0.3514 | 0.9118 | 13 | 24 | 6 | 62 |
| Product QK | 3 | 0.8079 | 0.7119 | 0.6576 | 0.4818 | 0.3243 | 0.9909 | 12 | 25 | 1 | 109 |
| Product QK | 4 | 0.7833 | 0.7443 | 0.6610 | 0.4925 | 0.3220 | 1.0000 | 19 | 40 | 0 | 122 |
| Product QK | 5 | 0.7961 | 0.6864 | 0.7631 | 0.5651 | 0.6122 | 0.9140 | 30 | 19 | 8 | 85 |
| IQP-ZZ QK | 1 | 0.5802 | 0.5017 | 0.5903 | 0.2065 | 0.3250 | 0.8557 | 13 | 27 | 14 | 83 |
| IQP-ZZ QK | 2 | 0.6475 | 0.5707 | 0.6328 | 0.3529 | 0.3243 | 0.9412 | 12 | 25 | 4 | 64 |
| IQP-ZZ QK | 3 | 0.8285 | 0.7409 | 0.6576 | 0.4818 | 0.3243 | 0.9909 | 12 | 25 | 1 | 109 |
| IQP-ZZ QK | 4 | 0.7002 | 0.6465 | 0.6186 | 0.4164 | 0.2373 | 1.0000 | 14 | 45 | 0 | 122 |
| IQP-ZZ QK | 5 | 0.8086 | 0.7677 | 0.7631 | 0.5651 | 0.6122 | 0.9140 | 30 | 19 | 8 | 85 |
| RBF | 1 | 0.5588 | 0.4967 | 0.6050 | 0.2282 | 0.3750 | 0.8351 | 15 | 25 | 16 | 81 |
| RBF | 2 | 0.6419 | 0.5604 | 0.6451 | 0.3530 | 0.3784 | 0.9118 | 14 | 23 | 6 | 62 |
| RBF | 3 | 0.7813 | 0.7000 | 0.6576 | 0.4818 | 0.3243 | 0.9909 | 12 | 25 | 1 | 109 |
| RBF | 4 | 0.7470 | 0.6960 | 0.6186 | 0.4164 | 0.2373 | 1.0000 | 14 | 45 | 0 | 122 |
| RBF | 5 | 0.8240 | 0.7491 | 0.7781 | 0.5836 | 0.6531 | 0.9032 | 32 | 17 | 9 | 84 |
| MLP | 1 | 0.4889 | 0.4055 | 0.5845 | 0.2829 | 0.2000 | 0.9691 | 8 | 32 | 3 | 94 |
| MLP | 2 | 0.6367 | 0.5206 | 0.6254 | 0.3252 | 0.3243 | 0.9265 | 12 | 25 | 5 | 63 |
| MLP | 3 | 0.7921 | 0.6586 | 0.6576 | 0.4818 | 0.3243 | 0.9909 | 12 | 25 | 1 | 109 |
| MLP | 4 | 0.6826 | 0.6281 | 0.6186 | 0.4164 | 0.2373 | 1.0000 | 14 | 45 | 0 | 122 |
| MLP | 5 | 0.7852 | 0.7678 | 0.7637 | 0.5817 | 0.5918 | 0.9355 | 29 | 20 | 6 | 87 |

TABLE S18
HISTORICAL HOLDOUT ($n = 181$) EVALUATED ONCE WITH THE FROZEN DEVELOPMENT-SELECTED THRESHOLDS.

| Model | AUROC | AP | BA | MCC | Sens. | Spec. | TP | FN | FP | TN |
|---|---|---|---|---|---|---|---|---|---|---|
| RF | 0.6926 | 0.5034 | 0.6175 | 0.3579 | 0.2653 | 0.9697 | 13 | 36 | 4 | 128 |
| LR | 0.6848 | 0.4907 | 0.5551 | 0.1777 | 0.1633 | 0.9470 | 8 | 41 | 7 | 125 |
| Product QK | 0.6974 | 0.4596 | 0.5332 | 0.1650 | 0.0816 | 0.9848 | 4 | 45 | 2 | 130 |
| IQP-ZZ QK | 0.6962 | 0.4633 | 0.5295 | 0.1357 | 0.0816 | 0.9773 | 4 | 45 | 3 | 129 |
| RBF | 0.7058 | 0.4992 | 0.5434 | 0.2002 | 0.1020 | 0.9848 | 5 | 44 | 2 | 130 |
| MLP | 0.4624 | 0.3024 | 0.5257 | 0.1110 | 0.0816 | 0.9697 | 4 | 45 | 4 | 128 |

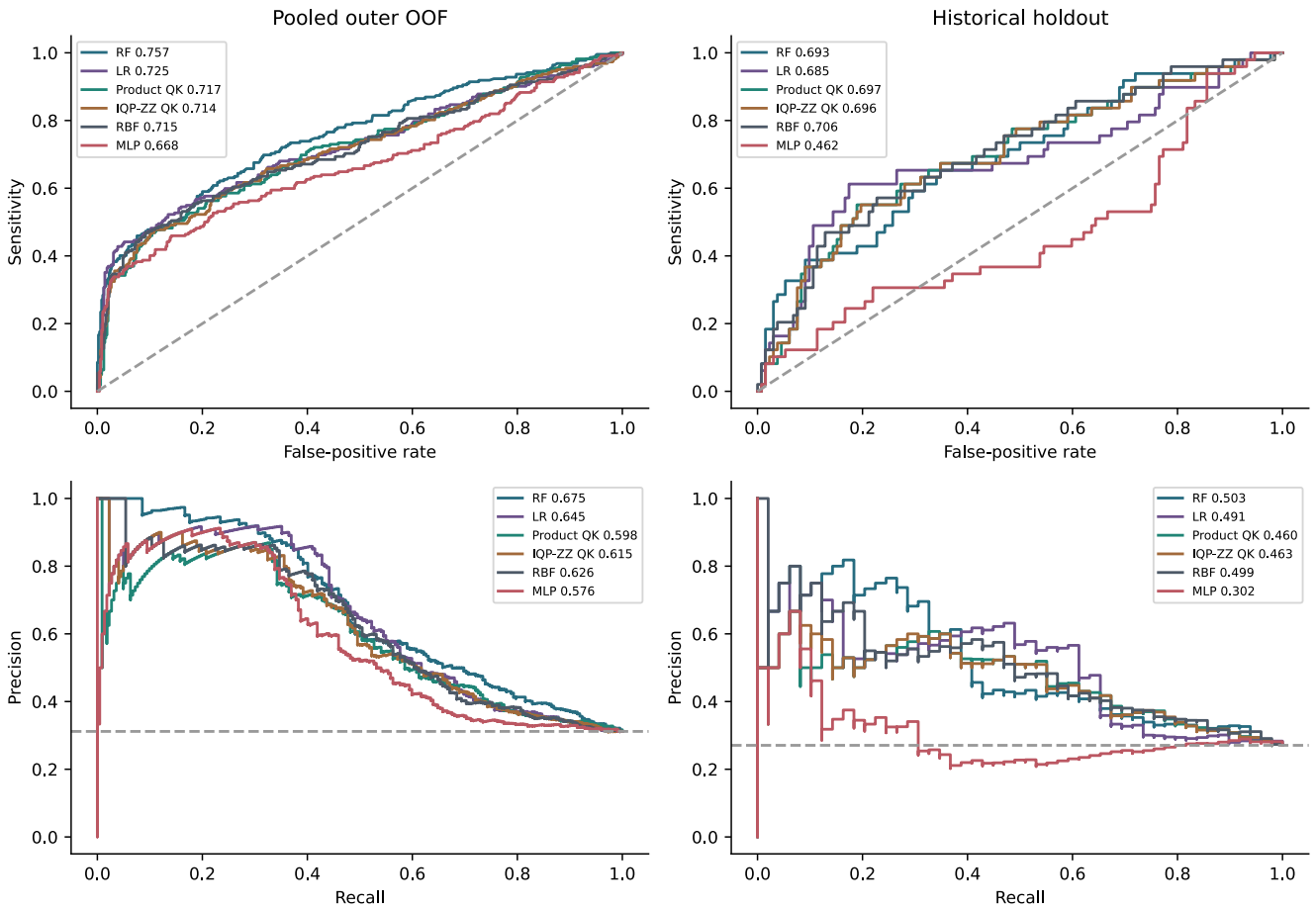



Fig. S9. ROC and precision–recall curves from archived pooled outer OOF scores and archived holdout scores. Legends show the corresponding pooled AUROC and AP. These curves neither adjust the frozen thresholds nor imply score calibration. Dashed horizontal AP references denote each partition's positive prevalence.

The frozen campaign identifies RF as the best mean outer-AUROC family, while the kernel differences are smaller than their fold dispersion. At final thresholds, the historical holdout detects only 4–13 of 49 toxic compounds across the six families, with 2–7 false positives among 132 non-toxic compounds. MLP holdout AUROC is 0.4624 and AP is 0.3024; its archived confusion matrix is TP=4, FN=45, FP=4, TN=128. The kernel holdout AUROCs are 0.7058 for RBF, 0.6974 for Product, and 0.6962 for IQP-ZZ.

TABLE S19
PAIRED AUROC DIFFERENCES ON IDENTICAL OUTER FOLDS, FOLLOWED BY A SEPARATE FROZEN-MODEL HOLDOUT CONTRAST.

| Contrast | F1 | F2 | F3 | F4 | F5 | Mean | Holdout |
|---|---|---|---|---|---|---|---|
| Product QK − RBF | +0.0031 | +0.0171 | +0.0265 | +0.0363 | -0.0279 | +0.0110 | -0.0083 |
| IQP-ZZ QK − RBF | +0.0214 | +0.0056 | +0.0472 | -0.0468 | -0.0154 | +0.0024 | -0.0096 |
| Product QK − IQP-ZZ QK | -0.0183 | +0.0115 | -0.0206 | +0.0831 | -0.0125 | +0.0086 | +0.0012 |

## S7. PAIRED CLUSTER BOOTSTRAP AND LABEL RANDOMIZATION

### A. *Paired cluster resampling*

The retained paired bootstrap samples 374 Development clusters with replacement and concatenates all member molecules, preserving a shared resample for both models in every contrast. The scores are the archived pooled nested OOF scores; predictive models are not retrained. For replicate $b$ and a ranking metric $M$,

$$\Delta_b = M(\mathbf{y}^{(b)}, \mathbf{f}_A^{(b)}) - M(\mathbf{y}^{(b)}, \mathbf{f}_B^{(b)}), \qquad I_{.95} = [Q_{.025}(\Delta_b), Q_{.975}(\Delta_b)]. \tag{S22}$$

The campaign uses 2,000 replicates and seed 20260824, skipping any replicate with only one class. The reported $P(\Delta > 0)$ is a bootstrap tail fraction rather than a null-calibrated significance-test $p$-value. The percentile intervals do not account for

repeating hyperparameter selection, threshold estimation, or model training, and therefore describe uncertainty conditional on the retained OOF scores.

TABLE S20
PAIRED CLUSTER BOOTSTRAP ON POOLED NESTED OOF SCORES, 2,000 REPLICATES. PERCENTILE INTERVALS ARE DESCRIPTIVE; TAIL FRACTIONS ARE NOT NULL-CALIBRATED HYPOTHESIS-TEST P-VALUES.

| Contrast | Metric | Observed $\Delta$ | Median | 95% low | High | $P(\Delta > 0)$ |
|---|---|---|---|---|---|---|
| RF − LR | AUROC | 0.0313 | 0.0311 | -0.0210 | 0.0932 | 0.8675 |
| RF − LR | AUPRC | 0.0295 | 0.0280 | -0.0235 | 0.0812 | 0.8520 |
| RF − MLP | AUROC | 0.0889 | 0.0879 | 0.0217 | 0.1616 | 0.9955 |
| RF − MLP | AUPRC | 0.0986 | 0.0958 | 0.0366 | 0.1522 | 0.9990 |
| Product QK − RBF | AUROC | 0.0017 | 0.0013 | -0.0197 | 0.0242 | 0.5410 |
| Product QK − RBF | AUPRC | -0.0281 | -0.0257 | -0.0536 | 0.0008 | 0.0330 |
| IQP-ZZ QK − RBF | AUROC | -0.0014 | -0.0011 | -0.0237 | 0.0255 | 0.4690 |
| IQP-ZZ QK − RBF | AUPRC | -0.0109 | -0.0088 | -0.0380 | 0.0210 | 0.2745 |
| Product QK − IQP-ZZ QK | AUROC | 0.0032 | 0.0018 | -0.0209 | 0.0279 | 0.5655 |
| Product QK − IQP-ZZ QK | AUPRC | -0.0172 | -0.0165 | -0.0591 | 0.0206 | 0.2080 |

Every reported quantum-versus-RBF interval includes zero. The point estimates based on pooled OOF ranking are distinct from the mean paired fold differences in Table S19; those values need not coincide.

### *B. Fixed-configuration Y-randomization*

The labels are globally permuted on Development 200 times, retaining each family's final selected configuration. Fits use the fixed outer partitions; kernel/scaler computations are reused where permitted. The full nested hyperparameter search is not repeated for each permutation. This is a fixed-configuration control [22].

TABLE S21
FIXED-CONFIGURATION LABEL RANDOMIZATION: 200 PERMUTATIONS. THE STORED EMPIRICAL FRACTION IS 0/200 FOR EACH FAMILY; THE PLUS-ONE FINITE-SIMULATION FRACTION IS 1/201.

| Model | Real pooled AUROC | Null mean | Null SD | Null P95 | Exceedances | Plus-one fraction |
|---|---|---|---|---|---|---|
| RF | 0.7565 | 0.4958 | 0.0301 | 0.5430 | 0 | 0.0050 |
| LR | 0.7252 | 0.4996 | 0.0310 | 0.5516 | 0 | 0.0050 |
| Product QK | 0.7171 | 0.4980 | 0.0304 | 0.5514 | 0 | 0.0050 |
| IQP-ZZ QK | 0.7140 | 0.4979 | 0.0303 | 0.5493 | 0 | 0.0050 |
| RBF | 0.7154 | 0.4978 | 0.0304 | 0.5471 | 0 | 0.0050 |
| MLP | 0.6677 | 0.5013 | 0.0256 | 0.5419 | 0 | 0.0050 |

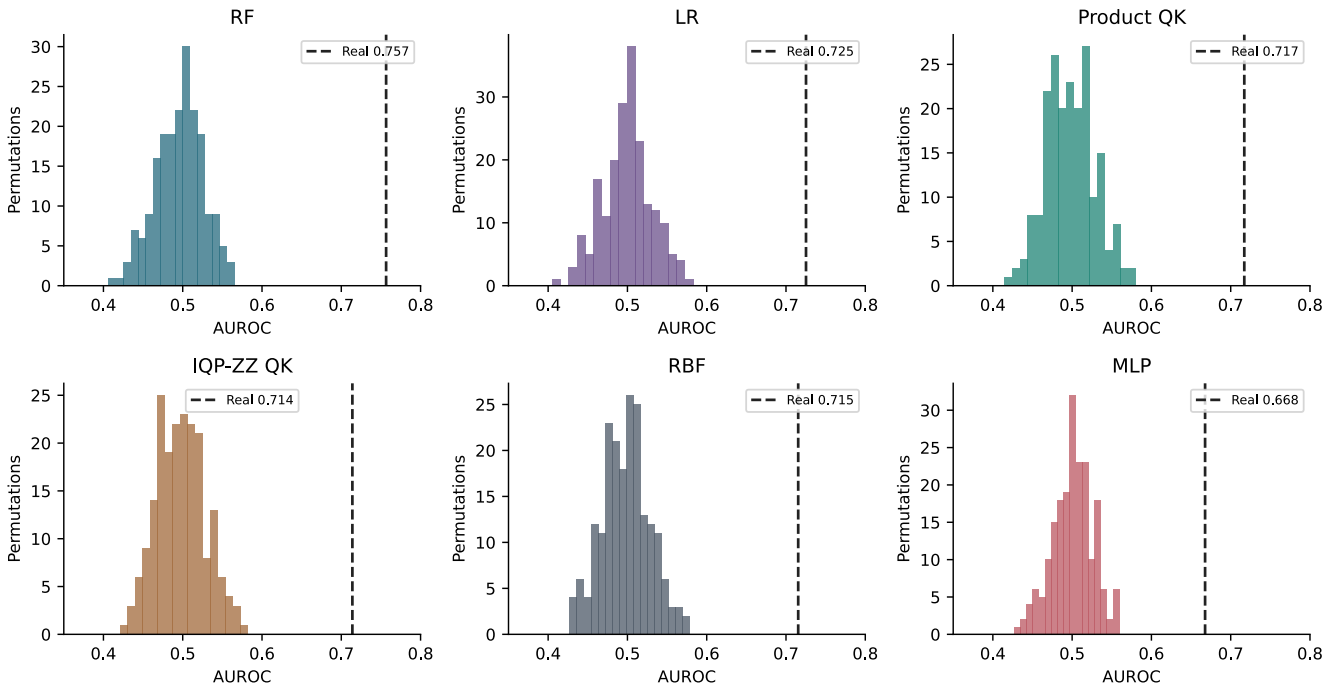



Fig. S10. Retained label-randomization AUROC distributions and the real pooled nested OOF AUROC for each model. The null fits use final fixed configurations, whereas the real OOF predictions use the nested selection procedure. This asymmetry limits a strict inferential interpretation.

No randomized score exceeds its real comparison score in the 200 retained replicates. The stored empirical fraction is zero; a finite-permutation plus-one summary is

$$p_{+1} = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{A_b \geq A_{\text{real}}\}}{B+1} = \frac{1}{201} \approx 0.004975. \tag{S23}$$

This expression avoids describing an empirical zero as a probability of zero. Because real and randomized fits use different selection histories and clusters are not permuted as intact labels, it is reported as a diagnostic tail fraction rather than a confirmatory test of the complete modeling pipeline.

## S8. Post Hoc Kernel Geometry and Bandwidth Controls

The final-selected kernels are reconstructed for all 712 Development molecules using one Development-fitted scaler. Labels are not used during reconstruction; their selected hyperparameters retain the preceding supervised-selection history. For $H = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$, centered alignment is

$$\text{CKA}(K, L) = \frac{\langle HKH, HLH \rangle_F}{\|HKH\|_F \|HLH\|_F}. \tag{S24}$$

This is the centered-kernel normalization used in alignment analysis [23]. The reported entropy-effective rank uses eigenvalues of the uncentered symmetric Gram matrix:

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \qquad r_{\text{eff}} = \exp\left( - \sum_{i:p_i>0} p_i \log p_i \right). \tag{S25}$$

Numerically negative eigenvalues caused by floating-point noise are clipped to zero for this supplementary calculation; this rank convention is separate from the minimum-eigenvalue QC tolerance.

TABLE S22

POST HOC UNCENTERED ENTROPY-EFFECTIVE RANK AND CENTERED ALIGNMENT ON ALL 712 DEVELOPMENT MOLECULES. A CONTROL RBF FIXES THE PRODUCT SMALL-ANGLE BANDWIDTH. NO LABELS ENTER MATRIX RECONSTRUCTION.

| Kernel | Effective rank | CKA vs. RBF(.03) | CKA vs. Product |
|---|---|---|---|
| RBF gamma=.03 | 5.517173 | 1.000000 | 0.982454 |
| Product s=.25 | 3.086433 | 0.982454 | 1.000000 |
| IQP-ZZ s=.125 | 3.156357 | 0.985072 | 0.999506 |
| RBF gamma=.015625 | 3.077806 | 0.983587 | 0.999794 |

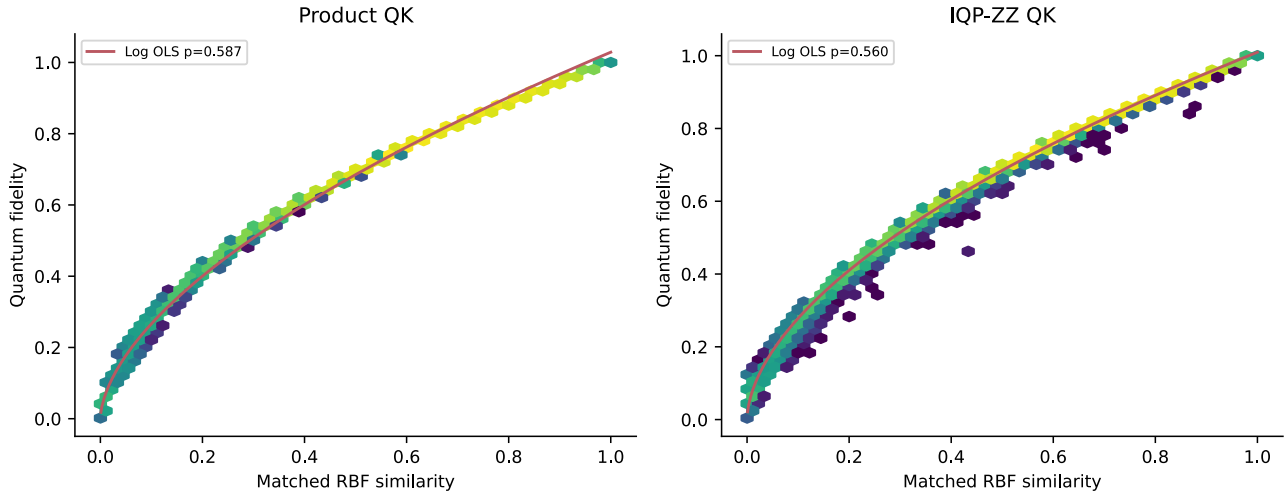



Fig. S11. All 253,116 unique off-diagonal Development molecular pairs. Hexagonal-bin colors summarize pair density; the red curve follows the stated unweighted log-space regression rather than an unspecified visual power fit. High CKA does not imply identical entries, identical spectra, or equal classification outcomes.

TABLE S23

EXACTLY SPECIFIED UNWEIGHTED FIT $\log K_Q = a + p \log K_{RBF}$ ON ALL 253,116 UNIQUE OFF-DIAGONAL PAIRS. ALSO SHOWN: FIT CONSTRAINED TO $a = 0$. PAIR DEPENDENCE PRECLUDES INTERPRETING ORDINARY REGRESSION ERRORS AS INDEPENDENT-PAIR UNCERTAINTY.

| Quantum kernel | Slope $p$ | Intercept $a$ | Origin-constrained $p_0$ |
|---|---|---|---|
| Product QK | 0.586551 | 0.028376 | 0.562509 |
| IQP-ZZ QK | 0.559648 | 0.009555 | 0.551553 |

The fit explicitly minimizes $\sum_{i<j}(\log K_{Q,ij} - a - p \log K_{R,ij})^2$; a second fit fixes $a = 0$. For Product's small-angle limit against the tuned RBF, the analytic power ratio is

$$p_{\text{local}} = \frac{\gamma_{\text{eff}}}{\gamma_{\text{RBF}}} = \frac{0.015625}{0.03} = 0.520833\ldots. \tag{S26}$$

The empirical fit covers all pairs, including pairs outside that local regime. The fitting convention and data range therefore affect the slope. The Product effective rank is close to that of the control RBF with $\gamma = 0.015625$, exposing the bandwidth contribution to the difference against the tuned RBF at $\gamma = 0.03$. The two quantum maps show almost unit alignment with each other under the selected scales.

## S9. APPLICABILITY DOMAINS, NOVELTY, AND CR8 SUPPORT

### A. *Development-only domain conventions*

The structural support statistic and descriptor support statistic are

$$S_{\max}(x) = \max_{j \in \mathcal{D}} T(x, x_j), \qquad d_5(x) = \frac{1}{5} \sum_{j \in \text{NN}_5(x;\mathcal{D})} \|\mathbf{z}(x) - \mathbf{z}_j\|_2, \tag{S27}$$

with the molecule itself excluded when $x \in \mathcal{D}$. Their cutoffs are the Development leave-one-out 5th percentile of $S_{\max}$ and 95th percentile of $d_5$:

$$\mathrm{IN}_S = \mathbb{1}\{S_{\max} \geq 0.223134\}, \qquad \mathrm{IN}_d = \mathbb{1}\{d_5 \leq 2.198975\}. \tag{S28}$$

The exact unrounded cutoffs are retained in the campaign JSON. Both percentiles are support conventions rather than performance-optimized thresholds, consistent with separating domain assessment from prediction validation [24].

TABLE S24
DESCRIPTOR SHIFT FOR THE HISTORICAL HOLDOUT. STANDARDIZED MEAN DIFFERENCE IS (HOLDOUT MEAN MINUS DEVELOPMENT MEAN)/POOLED SAMPLE SD.

| Descriptor | Development median | Holdout median | Std. mean difference |
|---|---|---|---|
| logP | 2.945 | 3.720 | 0.1847 |
| MW | 303.581 | 331.412 | 0.1731 |
| TPSA | 65.800 | 58.200 | -0.0469 |
| HBD | 1.000 | 1.000 | 0.1202 |
| RotB | 4.000 | 4.000 | 0.0214 |
| ArR | 1.000 | 1.000 | 0.1567 |
| nHal | 0.000 | 1.000 | 0.3478 |
| nOP | 0.000 | 0.000 | -0.2188 |
| LiPHEX | 0.202 | 0.227 | -0.0273 |
| fSASA | 0.154 | 0.139 | -0.1053 |

TABLE S25
STRUCTURAL AND DESCRIPTOR NEIGHBORHOODS RELATIVE TO DEVELOPMENT. DEVELOPMENT SUMMARIES EXCLUDE THE MOLECULE ITSELF.

| Partition | Mean $S_{max}$ | Median | P05 | Mean $d_5$ | Median $d_5$ | Both IN |
|---|---|---|---|---|---|---|
| Development LOO | 0.5102 | 0.5122 | 0.2231 | 1.1914 | 1.0553 | 650 |
| Holdout | 0.4046 | 0.3710 | 0.2115 | 1.3900 | 1.2029 | 164 |

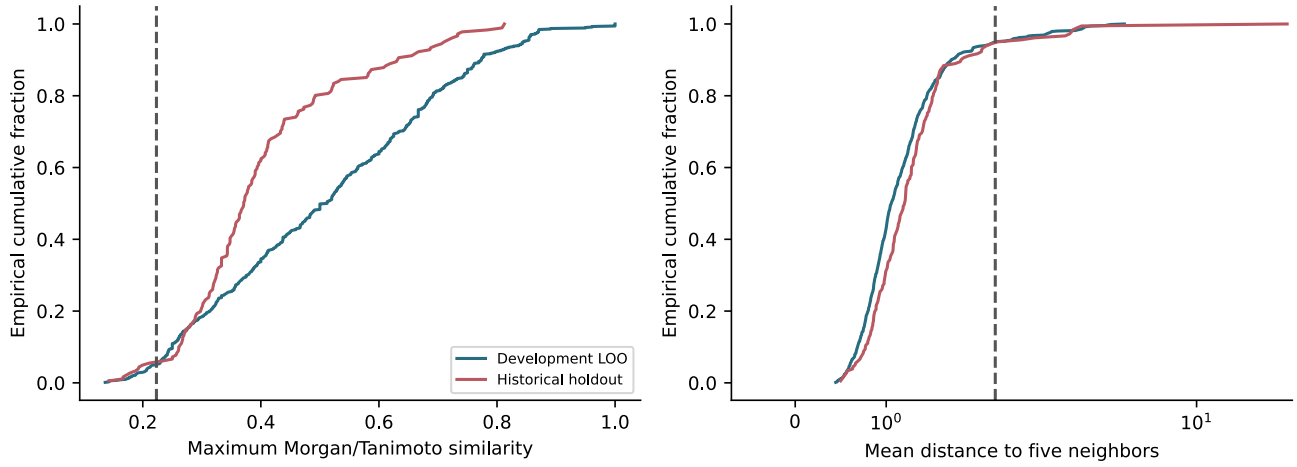



Fig. S12. Neighborhood novelty relative to Development: maximum structural similarity and mean descriptor distance to five neighbors. The descriptor-distance axis uses a symmetric logarithmic scale beyond its linear central region so extreme molecules remain visible.

## B. Complete external decisions

TABLE S26
FROZEN CR8 DUAL APPLICABILITY DOMAIN. STRUCTURAL CUTOFF 0.223134; DESCRIPTOR CUTOFF 2.198975.

| Compound | Class | $S_{max}$ | Struct. | $d_5$ | Desc. | Dual |
|---|---|---|---|---|---|---|
| Isotianil | 0 | 0.2642 | IN | 1.4041 | IN | IN/IN |
| Fluoxapiprolin | 0 | 0.5408 | IN | 2.4376 | OUT | IN/OUT |
| Ethiprole | 1 | 0.7400 | IN | 1.7519 | IN | IN/IN |
| Cyclobutrifluram | 0 | 0.2647 | IN | 1.0123 | IN | IN/IN |
| Isoprothiolane | 0 | 0.2973 | IN | 1.3682 | IN | IN/IN |
| Isocycloseram | 1 | 0.2326 | IN | 1.5794 | IN | IN/IN |
| Spiropidion | 0 | 0.3662 | IN | 1.1636 | IN | IN/IN |
| Florylpicoxamid | 0 | 0.3085 | IN | 1.6809 | IN | IN/IN |

TABLE S27
EVERY FROZEN CR8 DECISION AND ITS SIGNED SCORE MARGIN $m = f - t$ IN PARENTHESES. MARGINS HAVE MODEL-SPECIFIC SCALES AND ARE NOT CALIBRATED PROBABILITIES.

| Compound | Class | RF | LR | Product QK | IQP-ZZ QK | RBF | MLP |
|---|---|---|---|---|---|---|---|
| Isotianil | 0 | 0 (-0.4602) | 0 (-0.2589) | 0 (-1.5826) | 0 (-1.3098) | 0 (-1.9031) | 0 (-0.0969) |
| Fluoxapiprolin | 0 | 0 (-0.2309) | 0 (-0.0631) | 0 (-0.6617) | 0 (-0.3681) | 0 (-0.8335) | 0 (-0.0374) |
| Ethiprole | 1 | 0 (-0.3305) | 0 (-0.1129) | 0 (-1.4426) | 0 (-1.1810) | 0 (-1.6829) | 0 (-0.0355) |
| Cyclobutrifluram | 0 | 0 (-0.4900) | 0 (-0.1235) | 0 (-1.0004) | 0 (-0.7366) | 0 (-1.1822) | 0 (-0.0538) |
| Isoprothiolane | 0 | 0 (-0.3937) | 0 (-0.1453) | 0 (-1.1431) | 0 (-0.8919) | 0 (-1.1812) | 0 (-0.1520) |
| Isocycloseram | 1 | 0 (-0.1922) | 0 (-0.0395) | 0 (-0.7675) | 0 (-0.4835) | 0 (-0.8865) | 0 (-0.0139) |
| Spiropidion | 0 | 0 (-0.3817) | 0 (-0.0238) | 0 (-1.0196) | 0 (-0.7708) | 0 (-1.1712) | 0 (-0.1095) |
| Florylpicoxamid | 0 | 0 (-0.1650) | 0 (-0.2379) | 0 (-1.1796) | 0 (-0.8951) | 0 (-1.3283) | 1 (+0.0107) |

TABLE S28
CR8 METRICS ARE DESCRIPTIVE ONLY: EIGHT COMPOUNDS AND TWO POSITIVES. NO PARAMETER OR THRESHOLD IS TUNED ON CR8.

| Model | AUROC | AP | BA | MCC | Sens. | Spec. | TP | FN | FP | TN |
|---|---|---|---|---|---|---|---|---|---|---|
| RF | 0.7500 | 0.5000 | 0.5000 | 0.0000 | 0.0000 | 1.0000 | 0 | 2 | 0 | 6 |
| LR | 0.7500 | 0.5000 | 0.5000 | 0.0000 | 0.0000 | 1.0000 | 0 | 2 | 0 | 6 |
| Product QK | 0.5000 | 0.3929 | 0.5000 | 0.0000 | 0.0000 | 1.0000 | 0 | 2 | 0 | 6 |
| IQP-ZZ QK | 0.5000 | 0.3929 | 0.5000 | 0.0000 | 0.0000 | 1.0000 | 0 | 2 | 0 | 6 |
| RBF | 0.5000 | 0.3929 | 0.5000 | 0.0000 | 0.0000 | 1.0000 | 0 | 2 | 0 | 6 |
| MLP | 0.8333 | 0.5833 | 0.4167 | -0.2182 | 0.0000 | 0.8333 | 0 | 2 | 1 | 5 |

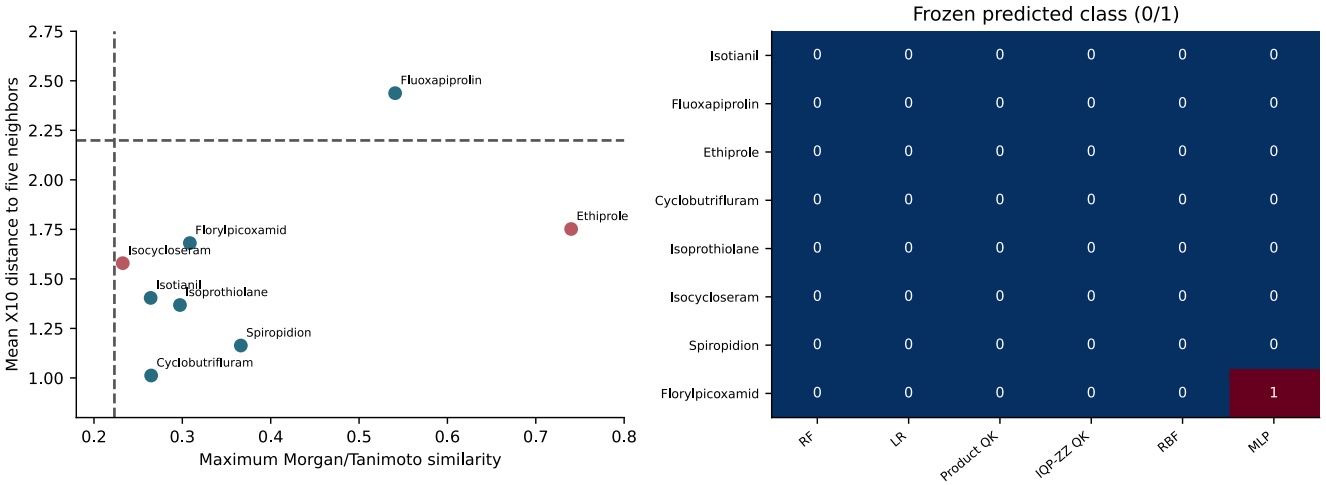



Fig. S13. Dual applicability domain and every frozen CR8 class prediction. Toxic compounds are red in the domain panel. Predicted classes in the matrix are the actual retained decisions, without retuning the thresholds.

Seven CR8 compounds are within both support domains. Fluoxapiprolin exceeds the descriptor-distance cutoff. Ethiprole and Isocycloseram lie within both domains but are false negatives for all six frozen models. MLP additionally produces one false positive, for Florylpicoxamid. With two positives, changing one decision would change sensitivity by 0.5; CR8 is therefore interpreted primarily at the molecule level. Domain membership quantifies proximity to Development and does not guarantee a correct prediction.

### *C. Class-specific and top-ten neighbors*

TABLE S29: Nearest Development neighbor in each class, separately for Morgan/Tanimoto structure and standardized X10 Euclidean distance; ties resolved by first input row. These are post hoc support diagnostics.

| CR8 compound | Class | Nearest by structure | $T$ | Nearest by X10 | $d$ |
|---|---|---|---|---|---|
| Isotianil | 0 | (alphaE)-2-[[6-(2-Cyanophenoxy)-4-pyrimidinyl]oxy]-alpha-(methoxymethylene)benzeneacetic acid methyl ester | 0.2464 | Benazolin (Ref: RD 7693) | 1.2357 |
| Isotianil | 1 | Fenpiclonil (Ref: CGA 142705) | 0.2642 | 1-[(6-Chloro-3-pyridinyl)methyl]-N-nitro-1H-imidazol-2-amine | 1.5495 |
| Fluoxapiprolin | 0 | 1-[4-[4-[5-(2,6-Difluorophenyl)-4,5-dihydro-3-isoxazolyl]-2-thiazolyl]-1-piperidinyl]-2-[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]ethanone | 0.5408 | 1-[4-[4-[5-(2,6-Difluorophenyl)-4,5-dihydro-3-isoxazolyl]-2-thiazolyl]-1-piperidinyl]-2-[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]ethanone | 2.1368 |
| Fluoxapiprolin | 1 | Cyclaniliprole (Ref: IKI-3106) | 0.1917 | Acrinathrin (Ref: RU 38702) | 2.3620 |
| Ethiprole | 0 | 2-Chloro-N-[2,6-dinitro-4-(trifluoromethyl)phenyl]-N-ethyl-6-fluorobenzenemethanamine | 0.2639 | Halauxifen (Ref: X11393729) | 1.4614 |
| Ethiprole | 1 | 5-Amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carbonitrile | 0.7400 | 5-Amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carbonitrile | 1.7065 |
| Cyclobutrifluram | 0 | Triflumuron (Ref: OMS 2015) | 0.2647 | N-[9-(Dichloromethylene)-1,2,3,4-tetrahydro-1,4-1H-pyrazole-4-carboxamide | 0.5551 |
| Cyclobutrifluram | 1 | Cyclaniliprole (Ref: IKI-3106) | 0.2644 | 4-Bromo-2-(4-chlorophenyl)-1-(ethoxymethyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile | 0.9281 |
| Isoprothiolane | 0 | N-Phenylcarbamic acid, 1-Methylethyl ester | 0.2973 | Nitrothal-isopropyl (Ref: BAS 300 OOF) | 1.2016 |
| Isoprothiolane | 1 | 2-[(Aminomethoxyphosphino thioyl)oxy]benzoic acid 1-methylethyl ester | 0.2292 | 3,3-Dimethyl-1-(methylthio)-2-butanone O-[(methylamino)carbonyl]oxime | 1.4374 |
| Isocycloseram | 0 | Chlozolinate (Ref: M 8164) | 0.2326 | 1-[4-[4-[5-(2,6-Difluorophenyl)-4,5-dihydro-3-isoxazolyl]-2-thiazolyl]-1-piperidinyl]-2-[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]ethanone | 1.5367 |
| Isocycloseram | 1 | 1-(3-Chloropyridin-2-yl)-N-[4-cyano-2-methyl-6-(methylcarbamoyl)phenyl]-3-[5-(trifluoromethyl)-2H-tetrazol-2-yl]methyl-1H-pyrazole-5-carboxamide | 0.2051 | Acrinathrin (Ref: RU 38702) | 1.5323 |
| Spiropidion | 0 | Carbonic acid, cis-3-(2,5-Dimethylphenyl)-8-methoxy-2-oxo-1-azaspiro[4.5]dec-3-en-4-yl ethyl ester | 0.3662 | Mefenpyr diethyl (Ref: HOE 107892) | 0.9132 |
| Spiropidion | 1 | Chlorobenzilate (Ref: ENT 18596) | 0.2667 | Rotenone (Ref: ENT 133) | 1.2803 |
| Florylpicoxamid | 0 | Fenpicoxamid (Ref: XDE 777) | 0.3085 | Propaquizafop (Ref: CGA 233380) | 1.2742 |
| Florylpicoxamid | 1 | Isoprocarb (Ref: OMS 32) | 0.2273 | Fluacrypyrim | 1.7756 |

The nearest toxic and nearest non-toxic neighbors are determined separately in structure space and in standardized X10 space. These post hoc comparisons do not alter the model, threshold, or representation. All top-ten structural neighbors are supplied in Table S30 and the campaign CSV, including their SMILES, labels, and local toxic fractions.

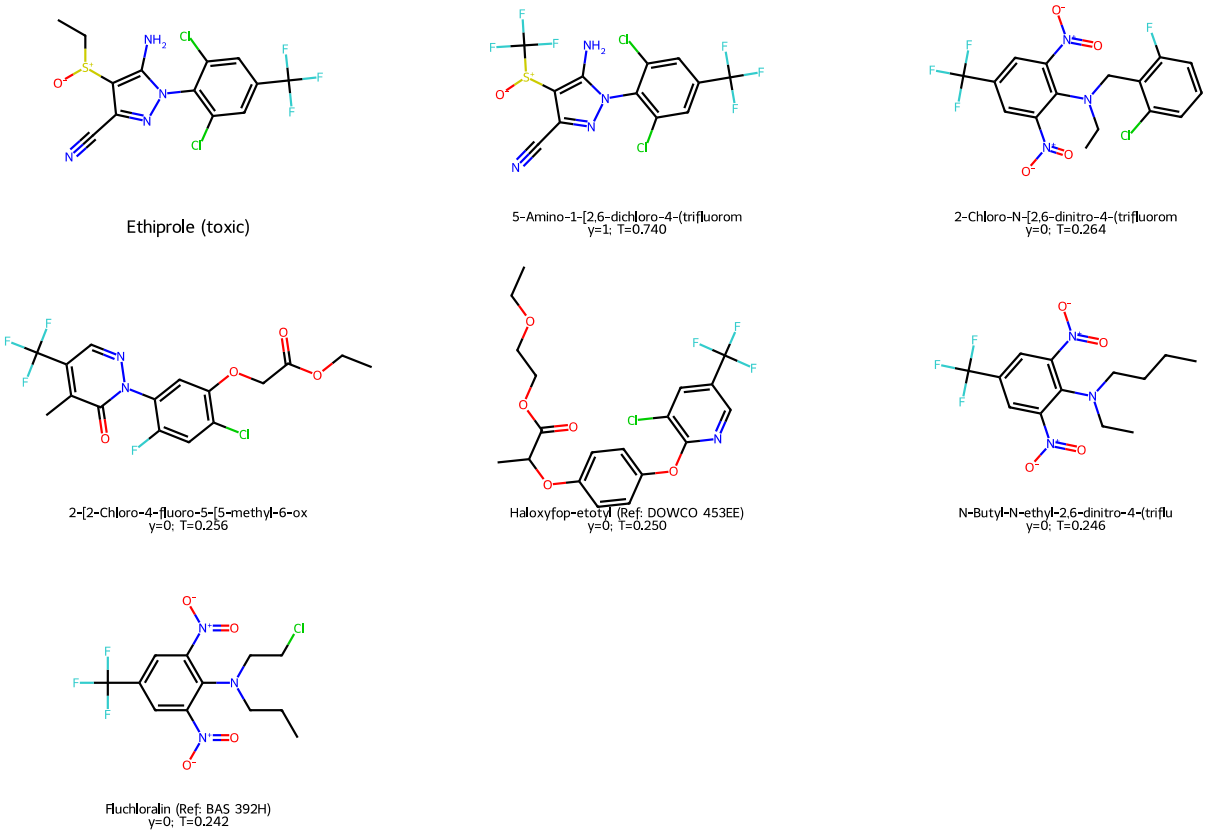



Fig. S14. Ethiprole and its first six retained Development structural neighbors. Neighbor labels and Morgan/Tanimoto similarities provide compound-specific context for the false negative.

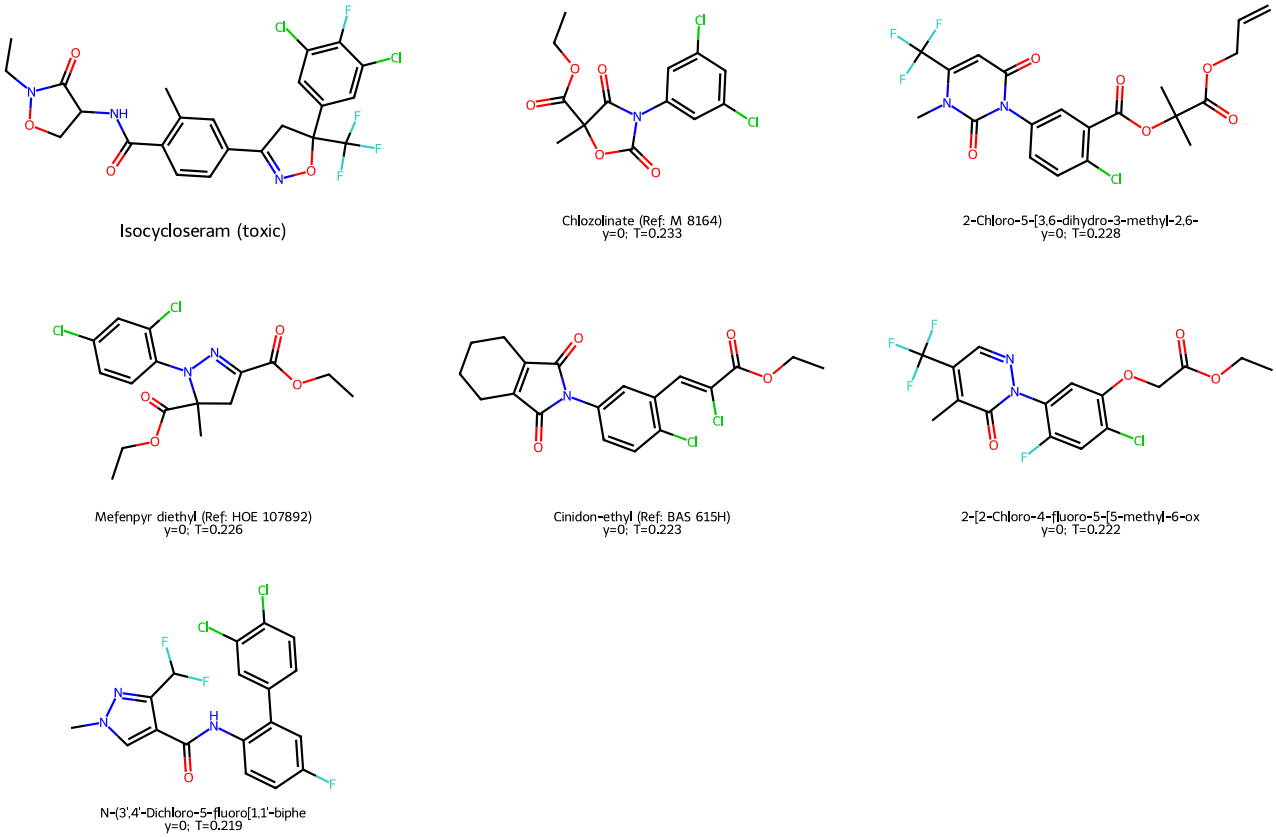



Fig. S15. Isocycloseram and its first six retained Development structural neighbors. Its comparatively low structural support differs from Ethiprole's support context. Neither comparison proves the toxicodynamic mechanism responsible for an error.

TABLE S30: Complete retained top-ten structural-neighbor records for all eight CR8 compounds (80 rows). Canonical source SMILES are recorded in the campaign neighbor CSV.

| CR8 compound | Rank | Development ID | Neighbor | Class | Tanimoto |
|---|---|---|---|---|---|
| Isotianil | 1 | 579 | Fenpiclonil (Ref: CGA 142705) | 1 | 0.2642 |
| Isotianil | 2 | 613 | (alphaE)-2-[[6-(2-Cyanophenoxy)-4-pyrimidinyl]oxy]-alpha-(methoxymethylene)benzeneacetic acid methyl ester | 0 | 0.2464 |
| Isotianil | 3 | 710 | Zarilamid (Ref: ICIA001) | 1 | 0.2373 |
| Isotianil | 4 | 298 | N-1H-Benzimidazol-2-yl carbamic acid methyl ester | 0 | 0.2321 |
| Isotianil | 5 | 992 | N-(Cyano-2-thienylmethyl)-4-ethyl-2-(ethylamino)-5-thiazolecarboxamide | 0 | 0.2286 |
| Isotianil | 6 | 536 | Halofenozide (Ref: RH 0345) | 0 | 0.2241 |
| Isotianil | 7 | 206 | Dichlobenil (Ref: H 133) | 1 | 0.2222 |
| Isotianil | 8 | 1028 | Penflufen (Ref: BYF-14182) | 0 | 0.2206 |
| Isotianil | 9 | 931 | Propyzamide (Ref: RH 315) | 0 | 0.2182 |
| Isotianil | 10 | 782 | Fluxapyroxad (Ref: BAS 700F) | 0 | 0.2174 |
| Fluoxapiprolin | 1 | 1023 | 1-[4-[4-[5-(2,6-Difluorophenyl)-4,5-dihydro-3-isoxazolyl]-2-thiazolyl]-1-piperidinyl]-2-[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]ethanone | 0 | 0.5408 |
| Fluoxapiprolin | 2 | 993 | 3-[2-Chloro-4-(methanesulfonyl)benzoyl]-4-(phenylsulfanyl)bicyclo[3.2.1]oct-3-en-2-one | 0 | 0.2336 |
| Fluoxapiprolin | 3 | 628 | 3-Chloro-2-[[(5-ethoxy-7-fluoro[1,2,4]triazolo[1,5-c]pyrimidin-2-yl)sulfonyl]amino]benzoic acid, Methyl ester | 0 | 0.2212 |
| Fluoxapiprolin | 4 | 797 | N-[9-(Dichloromethylene)-1,2,3,4-tetrahydro-1,4-1H-pyrazole-4-carboxamide | 0 | 0.2182 |
| Fluoxapiprolin | 5 | 301 | Benazolin (Ref: RD 7693) | 0 | 0.2128 |
| Fluoxapiprolin | 6 | 356 | Piperalin (Ref: HSDB 7058) | 0 | 0.2115 |

*Continued on next page*

*Table S30 continued*

| CR8 compound | Rank | Development ID | Neighbor | Class | Tanimoto |
|---|---|---|---|---|---|
| Fluoxapiprolin | 7 | 649 | 2-[2-Chloro-5-[4-chloro-5-(difluoromethoxy)-1-methyl-1H-pyrazol-3-yl]-4-fluorophenoxy]acetic acid ethyl ester | 0 | 0.2072 |
| Fluoxapiprolin | 8 | 779 | N-(3',4'-Dichloro-5-fluoro[1,1'-biphenyl]-2-yl)-3-(difluoromethyl)-1-methyl-1H-pyrazole-4-carboxamide | 0 | 0.2037 |
| Fluoxapiprolin | 9 | 791 | N-2-[(4,6-Dimethoxypyrimidin-2-yl)(hydroxy)methyl]-6-(methoxymethyl)phenyl-1,1-difluoromethanesulfonamide | 0 | 0.2037 |
| Fluoxapiprolin | 10 | 527 | Bensulfuron (Ref: IN R9419) | 0 | 0.2000 |
| Ethiprole | 1 | 387 | 5-Amino-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(trifluoromethyl)sulfinyl]-1H-pyrazole-3-carbonitrile | 1 | 0.7400 |
| Ethiprole | 2 | 526 | 2-Chloro-N-[2,6-dinitro-4-(trifluoromethyl)phenyl]-N-ethyl-6-fluorobenzenemethanamine | 0 | 0.2639 |
| Ethiprole | 3 | 696 | 2-[2-Chloro-4-fluoro-5-[5-methyl-6-oxo-4-(trifluoromethyl)-1(6H)-pyridazinyl]phenoxy]acetic acid, Ethyl ester | 0 | 0.2564 |
| Ethiprole | 4 | 743 | Haloxyfop-etotyl (Ref: DOWCO 453EE) | 0 | 0.2500 |
| Ethiprole | 5 | 34 | N-Butyl-N-ethyl-2,6-dinitro-4-(trifluoromethyl)benzenamine | 0 | 0.2462 |
| Ethiprole | 6 | 359 | Fluchloralin (Ref: BAS 392H) | 0 | 0.2424 |
| Ethiprole | 7 | 241 | 2,6-Dinitro-N,N-dipropyl-4-(trifluoromethyl)benzenamine | 0 | 0.2419 |
| Ethiprole | 8 | 378 | N-Ethyl-N-(2-methyl-2-propenyl)-2,6-dinitro-4-(trifluoromethyl)benzenamine | 0 | 0.2388 |
| Ethiprole | 9 | 400 | 2-Chloro-1-(3-ethoxy-4-nitrophenoxy)-4-(trifluoromethyl)benzene | 0 | 0.2329 |
| Ethiprole | 10 | 939 | 4-Bromo-2-(4-chlorophenyl)-1-(ethoxymethyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile | 1 | 0.2192 |
| Cyclobutrifluram | 1 | 423 | Triflumuron (Ref: OMS 2015) | 0 | 0.2647 |
| Cyclobutrifluram | 2 | 805 | Cyclaniliprole (Ref: IKI-3106) | 1 | 0.2644 |
| Cyclobutrifluram | 3 | 1025 | 1-(3-Chloropyridin-2-yl)-N-[4-cyano-2-methyl-6-(methylcarbamoyl)phenyl]-3-[5-(trifluoromethyl)-2H-tetrazol-2-yl]methyl-1H-pyrazole-5-carboxamide | 1 | 0.2604 |
| Cyclobutrifluram | 4 | 544 | Chlorantraniliprole (Ref: DPX E2Y45) | 1 | 0.2439 |
| Cyclobutrifluram | 5 | 777 | 2-[1-[2-[[(3,5-Difluorophenyl)amino]carbonyl]hydrazinylidene]ethyl]-3-pyridinecarboxylic acid | 0 | 0.2432 |
| Cyclobutrifluram | 6 | 525 | Flutolanil (Ref: NNF 136) | 0 | 0.2429 |
| Cyclobutrifluram | 7 | 680 | Picolinafen (Ref: BAS 700H) | 0 | 0.2400 |
| Cyclobutrifluram | 8 | 722 | Pyrifenox (Ref: CGA 179945) | 0 | 0.2394 |
| Cyclobutrifluram | 9 | 771 | Isopyrazam (Ref: SYN 520453) | 0 | 0.2375 |
| Cyclobutrifluram | 10 | 178 | N-(3-Chlorophenyl)carbamic acid, 1-Methylethyl ester | 0 | 0.2344 |
| Isoprothiolane | 1 | 134 | N-Phenylcarbamic acid, 1-Methylethyl ester | 0 | 0.2973 |
| Isoprothiolane | 2 | 265 | Bromopropylate (Ref: GS 19851) | 0 | 0.2821 |
| Isoprothiolane | 3 | 953 | Nitrothal-isopropyl (Ref: BAS 300 OOF) | 0 | 0.2750 |
| Isoprothiolane | 4 | 178 | N-(3-Chlorophenyl)carbamic acid, 1-Methylethyl ester | 0 | 0.2558 |
| Isoprothiolane | 5 | 498 | Diethofencarb (Ref: S 1605) | 0 | 0.2391 |
| Isoprothiolane | 6 | 709 | Iprovalicarb (Ref: SZX 0722) | 0 | 0.2340 |
| Isoprothiolane | 7 | 153 | Dinobuton (Ref: OMS 1056) | 0 | 0.2308 |
| Isoprothiolane | 8 | 455 | 2-[(Aminomethoxyphosphinothioyl)oxy]benzoic acid 1-methylethyl ester | 1 | 0.2292 |
| Isoprothiolane | 9 | 503 | Isofenphos-methyl | 1 | 0.2157 |
| Isoprothiolane | 10 | 686 | 2-(4-Methoxy[1,1'-biphenyl]-3-yl)hydrazinecarboxylic acid, 1-Methylethyl ester | 1 | 0.2157 |
| Isocycloseram | 1 | 456 | Chlozolinate (Ref: M 8164) | 0 | 0.2326 |
| Isocycloseram | 2 | 998 | 2-Chloro-5-[3,6-dihydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)-1(2H)-pyrimidinyl]benzoic acid 1,1-dimethyl-2-oxo-2-(2-propenyloxy)ethyl ester | 0 | 0.2277 |
| Isocycloseram | 3 | 704 | Mefenpyr diethyl (Ref: HOE 107892) | 0 | 0.2258 |
| Isocycloseram | 4 | 654 | Cinidon-ethyl (Ref: BAS 615H) | 0 | 0.2234 |
| Isocycloseram | 5 | 696 | 2-[2-Chloro-4-fluoro-5-[5-methyl-6-oxo-4-(trifluoromethyl)-1(6H)-pyridazinyl]phenoxy]acetic acid, Ethyl ester | 0 | 0.2222 |
| Isocycloseram | 6 | 779 | N-(3',4'-Dichloro-5-fluoro[1,1'-biphenyl]-2-yl)-3-(difluoromethyl)-1-methyl-1H-pyrazole-4-carboxamide | 0 | 0.2188 |
| Isocycloseram | 7 | 450 | Dimefuron (Ref: RP 23465) | 0 | 0.2128 |
| Isocycloseram | 8 | 202 | Methazole (Ref: VCS-438) | 0 | 0.2099 |
| Isocycloseram | 9 | 642 | Flupoxam (Ref: MON 18500) | 0 | 0.2079 |
| Isocycloseram | 10 | 598 | alpha,2-Dichloro-5-[4-(difluoromethyl)-4,5-dihydro-3-methyl-5-oxo-1H-1,2,4-triazol-1-yl]-4-fluorobenzenepropanoic acid, Ethyl ester | 0 | 0.2062 |
| Spiropidion | 1 | 762 | Carbonic acid, cis-3-(2,5-Dimethylphenyl)-8-methoxy-2-oxo-1-azaspiro[4.5]dec-3-en-4-yl ethyl ester | 0 | 0.3662 |

*Continued on next page*

*Table S30 continued*

| CR8 compound | Rank | Development ID | Neighbor | Class | Tanimoto |
|---|---|---|---|---|---|
| Spiropidion | 2 | 456 | Chlozolinate (Ref: M 8164) | 0 | 0.3231 |
| Spiropidion | 3 | 704 | Mefenpyr diethyl (Ref: HOE 107892) | 0 | 0.2703 |
| Spiropidion | 4 | 51 | Chlorobenzilate (Ref: ENT 18596) | 1 | 0.2667 |
| Spiropidion | 5 | 654 | Cinidon-ethyl (Ref: BAS 615H) | 0 | 0.2667 |
| Spiropidion | 6 | 406 | Procymidone (Ref: S 7131) | 0 | 0.2373 |
| Spiropidion | 7 | 649 | 2-[2-Chloro-5-[4-chloro-5-(difluoromethoxy)-1-methyl-1H-pyrazol-3-yl]-4-fluorophenoxy]acetic acid ethyl ester | 0 | 0.2317 |
| Spiropidion | 8 | 598 | alpha,2-Dichloro-5-[4-(difluoromethyl)-4,5-dihydro-3-methyl-5-oxo-1H-1,2,4-triazol-1-yl]-4-fluorobenzenepropanoic acid, Ethyl ester | 0 | 0.2278 |
| Spiropidion | 9 | 356 | Piperalin (Ref: HSDB 7058) | 0 | 0.2237 |
| Spiropidion | 10 | 656 | 2,2-Dimethylpropanoic acid 8-(2,6-diethyl-4-methylphenyl)-1,2,4,5- tetrahydro-7-oxo-7H-pyrazolo[1,2-d][1,4,5]oxadiazepin-9-yl ester | 0 | 0.2237 |
| Florylpicoxamid | 1 | 1032 | Fenpicoxamid (Ref: XDE 777) | 0 | 0.3085 |
| Florylpicoxamid | 2 | 650 | Benthiavalicarb-isopropyl (Ref: KIF-230) | 0 | 0.2317 |
| Florylpicoxamid | 3 | 415 | Isoprocarb (Ref: OMS 32) | 1 | 0.2273 |
| Florylpicoxamid | 4 | 581 | N-(4-Fluorophenyl)-N-(1-methylethyl)-2-[[5-(trifluoromethyl)-1,3,4-thiadiazol-2-yl]oxy]acetamide | 0 | 0.2250 |
| Florylpicoxamid | 5 | 709 | Iprovalicarb (Ref: SZX 0722) | 0 | 0.2222 |
| Florylpicoxamid | 6 | 455 | 2-[(Aminomethoxyphosphinothioyl)oxy]benzoic acid 1-methylethyl ester | 1 | 0.2192 |
| Florylpicoxamid | 7 | 822 | Halauxifen (Ref: X11393729) | 0 | 0.2179 |
| Florylpicoxamid | 8 | 777 | 2-[1-[2-[[(3,5-Difluorophenyl)amino]carbonyl]hydrazinylidene]ethyl]-3-pyridinecarboxylic acid | 0 | 0.2125 |
| Florylpicoxamid | 9 | 995 | Benthiavalicarb | 0 | 0.2125 |
| Florylpicoxamid | 10 | 825 | Benzyl 4-amino-3-chloro-6-(4-chloro-2-fluoro-3-methoxyphenyl)-5-fluoropyridine-2-carboxylate | 0 | 0.2118 |

## S10. Exploratory Development History and Analysis Boundaries

### A. Representation screening recorded in the exploratory record

The Phase 0 and Phase 1 notebooks establish descriptive findings and a leakage-safe preprocessing contract. The exploratory record also documents earlier representation experiments summarized below. These values belong to exploratory pre-screening, use a different data version or model lineup, and are not substituted for the frozen nested results in Section S6.

TABLE S31
Exploratory history from the X10 development record. Approximate signs reflect the source record. These experiments are separate from the final six-family campaign.

| Representation | Model / analysis | Recorded result |
|---|---|---|
| X5 in-house | RBF | Pooled OOF AUROC $\sim 0.522$. |
| X7 in-house | RBF | Pooled OOF AUROC $\sim 0.604$. |
| FREE20 | CatBoost | OOF AUROC 0.6925, AP 0.5601, MCC 0.2717. |
| FREE20 + LiPHEX | CatBoost | OOF AUROC $\sim 0.6917$; inconclusive incremental change. |
| X10, remove HBD | LR / RBF / CatBoost | Development OOF AUROC changes +0.0142 / +0.0125 / +0.0074. |
| X10 versus X9 | Historical holdout LR | AUROC 0.6891 versus 0.6877. |
| X10 versus X9 | Historical holdout RBF | AUROC 0.6977 versus 0.6981. |
| X10 versus X9 | Historical holdout CatBoost | AUROC 0.7031 versus 0.6869. |

These earlier ablations motivated retaining X10 and declaring candidate ablations rather than silently replacing the primary feature set. The uploaded final campaign contains no complete nested descriptor-ablation result matrix; newly computed predictive ablation values are therefore not asserted here. The descriptive nOP subgroup and purity analyses are not predictive ablations.

### B. Random-split references and identifier status

The notebooks include code for alternative partition construction, including a 625/179/89 split. This partition uses a different protocol and is not the 712/181/8 campaign. The retained artifact set does not establish a completed, matched random-split comparison using the same frozen inputs, six families, and nested threshold procedure. Such a result cannot be recovered by relabeling the alternative split.

The earlier notebook reports duplicate CID mappings, while molecular IDs and X10 coordinates remain the computational keys. CID/CAS/name fields are excluded from modeling. The supplementary checks validate the actual input hashes and cluster partitioning rather than using an identifier field as a substitute for molecular similarity.

## S11. Reproducibility and Remaining Evidence Limits

Predictive computations are represented by the original retained campaign CSVs rather than new model fits. Supplementary checks verified every archived threshold, margin, confusion count, AUROC, AP, BA, and MCC from the stored predictions, and reproduced both applicability-domain thresholds and all eight external-domain coordinates. Formula and implementation checks are reported in Section S4. The main frozen campaign source is available at https://github.com/andresquesadag/BeeQ [25]. The campaign seed is 20260824; supplementary descriptor/embedding diagnostics use seed 42. Frozen outer folds are read from the CSVs and never regenerated. The recorded original environment uses Python 3.12.8, NumPy 2.5.1, pandas 2.3.3, scikit-learn 1.9.0, RDKit 2026.03.5, matplotlib 3.11.1, and Qiskit 2.5.2. The campaign environment record reports Git commit `eeac3bf14f3e90f29494c3eb1c9837363fa88142` with a dirty working tree. That commit alone therefore does not identify every local source byte; the campaign source snapshot and its provenance JSON provide the additional campaign provenance.

TABLE S32
CANONICAL INPUT INTEGRITY. FULL SHA-256 STRINGS ARE RECORDED IN THE FROZEN CAMPAIGN CONFIGURATION AND INPUT-HASH CSV; ALL FOUR MATCH THE INPUTS USED HERE.

| Input | SHA-256 prefix | Verification |
|---|---|---|
| `master_RDKitFixed.csv` | `a0a6177b3319dba8` | Match |
| `train_RDKitFixed.csv` | `06a0817c082d7715` | Match |
| `test_RDKitFixed.csv` | `3f964207e5d73231` | Match |
| `ExternalFinal_RDKitFixed.csv` | `70a01eedd970991e` | Match |

The 51-entry original campaign manifest is also audited against the campaign archive. Thirteen entries match byte-for-byte. The remaining 38 are text files whose recorded hashes are recovered exactly by restoring CRLF line endings; there are no unexplained mismatches. This export normalization does not change numerical CSV contents. All 51 entries are therefore accounted for after text normalization.

The scientifically material limitations are the historical status of the holdout, small CR8 size, global label permutations with fixed selected configurations, conditional bootstrap intervals, lack of noisy/hardware quantum evaluation, and incomplete upstream descriptor/curation and quantitative CR8 endpoint provenance. A complete curation ledger, matched predictive descriptor ablations, a matched random-split baseline, and quantitative CR8 $LD_{50}$ source records were not available for this evaluation.

## S12. Related Work

In order to find previous related work, we conducted a literature review, using ApisTox benchmark paper [1] as a starting point, which supplied an initial set of keywords through its own reference list. This set was extended with keywords from exploratory searches on chemical toxicity, drug discovery, machine learning (ML), and quantum machine learning (QML). From this pool, three Scopus query strings were built: one for honey bee toxicity classification with ML, one for molecular toxicity prediction with ML, and one for molecular property or toxicity prediction with QML [25].

Fig. S16 shows the resulting pipeline. The three queries were merged and deduplicated, then screened by title, abstract, and keywords, since full text was not read at this stage. The remaining records were scored under a five-criterion rubric spanning endpoint relevance, method relevance, dataset comparability, evaluation strength, and novelty importance, and the highest-ranked records were added to a reference manager for full-text retrieval. Each of these records was then read in full, together with the ApisTox paper that had motivated the initial keyword search but was not itself returned by the queries.

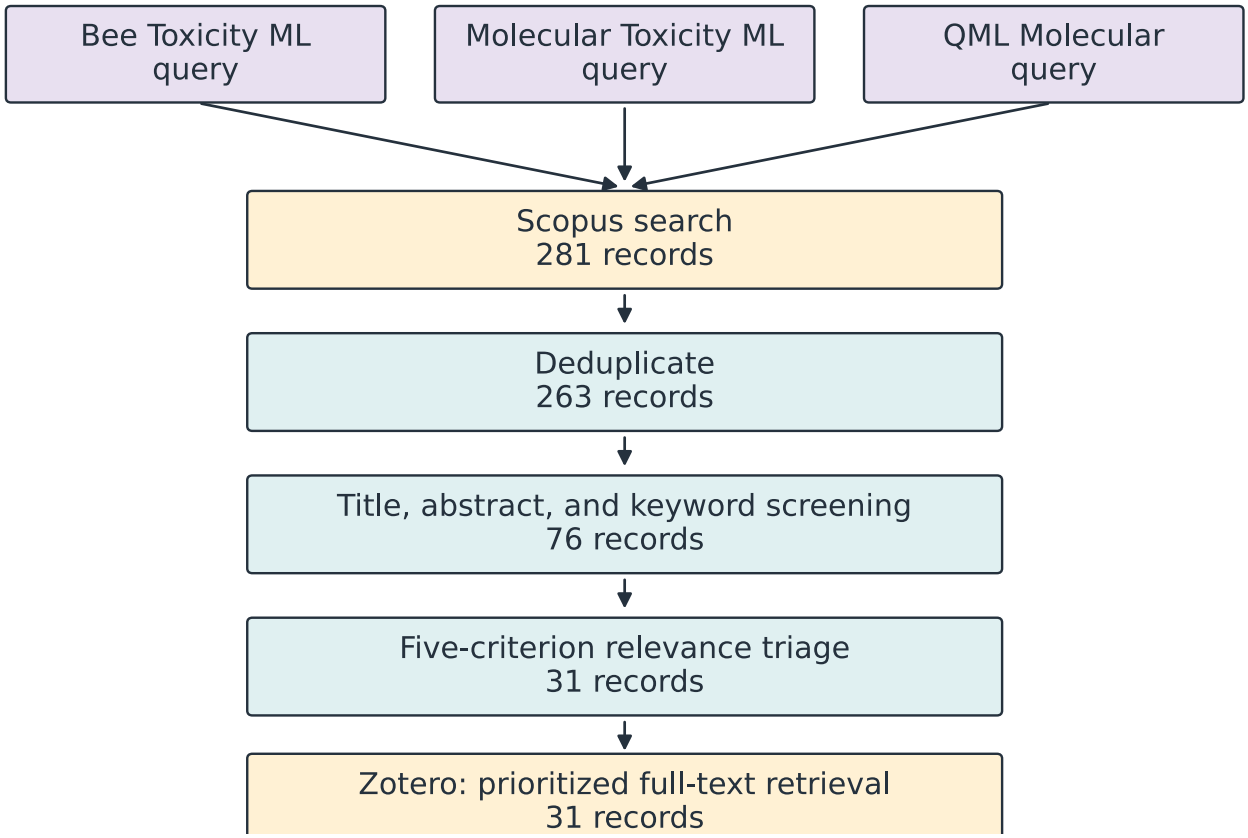



Fig. S16. Literature search and screening pipeline, from the three Scopus query strings to the final list of prioritized records forwarded to a reference manager for full-text retrieval.

### A. *Classical Honey-Bee Toxicity Modeling*

Previous classical computing approaches classify acute honey-bee toxicity from its molecular structure using different representations for each molecule. In [26] they calculated two-dimensional Dragon descriptors and then made a narrowed-down selection of those molecular descriptors using genetic algorithm and random forest selection, to then fit those descriptors into Random Forest (RF) and k-Nearest Neighbour (kNN) classifiers. Molecular descriptors were also used as the input representation in [27], where they evaluated two large descriptor pools, with 319 MOE and 353 ADMET Predictor descriptors, which were separately reduced through descriptor selection before training ANN groups. Another approach was used in [28]. They measured the similarity between molecular graphs using a fixed-length random walk graph kernel and then used the resulting Gram matrix as input to an SVM. In later work, molecular fingerprints were used as a representation. In [29] they crossed nine fingerprint families with six learning algorithms, including Naive Bayes(NB), Artificial Neural Network (ANN), C4.5, Support Vector Machine (SVM), RF and kNN. Producing 54 binary classification models. One of the fingerprint families

that [29] included was MACCS, which was also included in [30] along with Morgan and FeatMorgan fingerprints, that were combined with SVM and RF classifiers.

The variety of representations is not about complexity or improving the results of their models, but rather because they are not solving the same classification problem, despite their research being on the same line. In [26], they separated classes at 100 $\mu$g/bee and modeled 411 compounds, while [27], [29], [30] used the 11 $\mu$g/bee regulatory threshold although their datasets were not equivalent. In [29] they modeled 676 structurally diverse pesticides for acute contact toxicity, while [30] built separate datasets for contact and oral toxicity, with 382 and 169 compounds, respectively. The contact dataset from BeeToxAI was later used by [28] for their graph-kernel comparison. [27] used a larger collection of 1,542 compounds by combining PPDB data with a proprietary agrochemical dataset. These differences in endpoint, exposure route, and chemical space make the reported performance between studies difficult to compare directly.

Furthermore, the studies differ in the assessment of the model's performance on novel molecules. [26] employed a MaxMin-type splitting by means of PubChem fingerprints and Tanimoto similarity to ensure structural dissimilarity between the training and test sets of compounds. The applicability domain was estimated using VEGA-HUB. Conversely, [29] utilized the random 80/20 split in combination with ten-fold cross-validation. In addition, they applied a kNN-distance approach, where performance of predictions on test compounds improves when the latter are limited to those supported by the training chemical space. [30] also included an applicability domain through the fingerprint-based approach and identified unreliable predictions in the BeeToxAI model via this approach. The applicability domain was estimated in [27] through the chemical clustering of data and further application of MACCS/Tanimoto similarity criterion as well as a descriptor-space-based criterion. Finally, [28] handled variability via performing the evaluation over 2,000 resampled runs despite the fact that no restriction of structural similarity is implied thereby.

These works provide several fundamental elements on which our research questions are based. First, [29], [30] prove the relevance of using SVM and RF in the classical model as a measure of acute honey bee toxicity, while [28] provides a direct precedent in applying kernel-based comparison of molecules in the context of SVM. Second, [26], [27] prove that the evaluation of the model must take into account the structural dissimilarity and the coverage of chemical space. Finally, the application of the descriptor selection by [26], [27] proves that the information about the molecule that is supplied to the model is crucial.

### *B. ApisTox, Structural Generalization, and Recent Models*

The introduction of ApisTox provided a common benchmark for studying high acute toxicity to honey bees in more challenging conditions. In particular, the dataset was created by Adamczyk et al., with 1,035 standardized and deduplicated compounds, of which 296 compounds were toxic, whereas 739 compounds were non-toxic based on the approximate 11 $\mu$g/bee toxicity threshold [1]. Moreover, the ApisTox dataset contains predefined MaxMin and approximate temporal partitions. Specifically, the MaxMin partition chooses the test set based on molecular similarity. On the other hand, the approximate temporal partition ensures that the test set is comprised of newer molecules.

Later, Adamczyk et al. performed experiments where ApisTox was used for evaluation of molecular fingerprints, graph kernels, Graph Neural Networks (GNNs), and pretrained molecular models using MaxMin and temporal partitions [31]. Ranking of the models was different depending on these experimental settings. In the MaxMin partitioning, the WL-OA graph kernel achieved an MCC value of 0.49 and an AUROC of 83.95%, whereas ECFP fingerprints showed MCC of 0.48 and AUROC of 78.16% in the temporal partition. GNNs and pretrained models were not superior to fingerprints and graph kernels consistently. Therefore, the study demonstrates that model complexity does not necessarily affect its performance if the test molecules belong to a different chemical space. Moreover, results concerning model performance can vary depending on the generalization that is needed in the split.

Recent work on the same problem highlight even more clearly how performance on prediction is different from generalization to structure. Specifically, Damião et al. used a GNN to train on the ApisTox dataset based on molecular graphs generated from SMILES [32]. Using the method of five-fold random cross-validation, they achieved 0.759 ± 0.030, 0.552 ± 0.057, and 0.763 ± 0.027 as average AUROC, MCC, and balanced accuracy for testing respectively. The training AUROC was around 0.97. It should be noted that the current study does not use MaxMin or temporal folds and only uses random folds.

The model BeeEcoTox is another example of such distinction [33]. Chen et al. utilized molecular graphs with the incorporation of semantic data from ChemFM along with other structural features for a set of about 1,139 agrochemicals. In its standard split setting, BeeEcoTox achieved an AUC of around 0.91 and recall of 0.90. In temporal-style evaluation, AUC dropped to approximately 0.72 and MCC to 0.22 when the distance between train-test chemical space grew. Such contrast demonstrates that good performance of a model in its standard split setup can be accompanied by poor performance in the newest part of chemical space. Overall, the ApisTox benchmark and these state-of-the-art models make structural generalization one of the crucial concerns for honey-bee toxicity prediction. It becomes clear that a comparison of model families should take into account not only their performance in random splits but also in lower structural space.

### C. Quantum Machine Learning for Molecular Prediction

Quantum machine learning (QML) has been employed in predicting properties and toxicities of molecules, but the studies in each domain are different from each other. According to Schuld and Killoran, QML was explained using quantum feature space, where a classical input $x$ was transformed to a quantum state $|\phi(x)\rangle$. In a supervised learning problem, Havlíček et al. used the concept of a quantum kernel estimator, which measures similarities among the samples and provides the computed kernel as input to the classical Support Vector Machine (SVM). For fidelity-based kernels, such a similarity can be represented as

$$K(x, x') = |\langle\phi(x)|\phi(x')\rangle|^2 \tag{S29}$$

This formulation is relevant to molecular prediction because the classifier can remain classical while the feature map determines a different similarity function between samples. [16], [17]

The applications of QML for toxicology and molecule-based data, however, are fewer. Suzuki and Katouda made use of parameterized quantum circuits for a quantitative structure-activity relationship model that deals with the toxicity of 221 phenols. The study made comparisons between the results obtained by their quantum models and those of multiple linear regression and radial basis function networks. [34]. Subsequently, Bhatia et al. investigated the pharmacokinetics and toxicology properties of small molecules using a quantum-kernel based framework with a Support Vector Classifier (SVC). Their work is more relevant to our scenario as it involves reduced molecular data, quantum feature representation, and kernel classifier as well as the comparison between the quantum and classical approaches. [35]

It is not necessarily the case that there will be a favorable comparison between classical and quantum models. In the paper of Harjanto et al., they encoded the data in material datasets into only four features and compared the classical SVM with QSVM, VQC, and QNN. Through repeated stratified evaluations, the classical SVM obtained the best classification results. It was observed by the researchers that the poor performance of the QML model can be attributed to the shortcomings of feature encoding, circuit expressibility, qubit number, and optimization. [36]

Therefore, the cited work gives examples of application of QML in toxicity prediction and quantum kernels in molecule classification; however, it lacks the case study of the honey bee toxicity problem. In particular, there was no comparison of the fidelity-based quantum kernel and classical kernel in ApisTox using the same representation, preprocessing, partitioning of the dataset taking into account molecular structure, and SVC formulation.

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